No.　41-32001-00301

Instruction Manual(Basic Operation)

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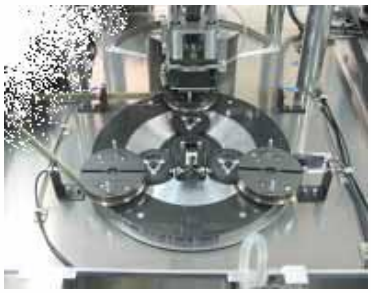
TIETECH

SAH-2200 Basic Operation Manual

Equipment name SAH-2200

Version 2.2ZU27



TIETECH

Thank you for selecting our handler.

Please read this manual carefully and understand thoroughly the precautions for use, outline of the equipment, operating instructions and matters concerning the safety when operating this equipment before carrying in, installing and operating this equipment, so that you can use this equipment safely for a long period of time. In the operation and maintenance manual for this equipment, the three hazard levels, i.e. Danger , Warning and Caution , are described. Please understand thoroughly these applicable descriptions before using this equipment.

"**Danger**" means that the inappropriate operation and maintenance or the wrongful use may result in serious injuries or even death of the operator or other personnel. the operator or other personnel must observe the precautions for risk prevention described here more strictly than the operation and maintenance procedures.
Be sure to operate the equipment in accordance with these precautions for risk prevention.

"**Warning**" means that failure to observe strictly the correct operating procedures or correct maintenance procedures may result in the serious injuries to the operator or other personnel. The precautions for hazard prevention are described here:
Be sure to operate the equipment in accordance with these precautions.

"**Caution**" means that failure to observe strictly the correct operating procedures or correct maintenance procedures may result in the injuries. The correct operating procedures and maintenance procedures are described here :
Be sure to operate the equipment correctly in accordance with these procedures.

Another indication includes:

Important : It describes the available and useful function of the equipment. examples of possible mistakes. and the precautions and necessary remedial measures to be taken; failure to observe these instructions may result in the damages to the equipment.

Remarks

- Illustrations and displays that appear in this manual may be different from those that appear in actual system due to special specifications required by customer, to which you are requested to understand beforehand.
- The contents of this manual are subject to change without notice for future upgrade.
- Duplication or transfer of this manual, part or whole, is strictly prohibited without our consent.

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1.In order to use this equipment safely

1) Electric shock hazard

This equipment requires grounding so that be sure to prepare the plug socket with the earth terminal and connect the earth line before using this equipment. Even when the main breaker is disconnected, the electricity is supplied to the main breaker. Be sure to disconnect the plug socket before conducting the maintenance work.

2) Electric shock hazard inside the equipment power source

Never touch the power source or the current-carrying parts such as noise filters. Before touching the terminal block during the maintenance work, it must be confirmed that the maintenance personnel(line keepers) have disconnected the power supply. Even after all the power supplies have been disconnected. The electricity may exist at some component parts inside the equipment, so that it must be confirmed using a tester that no electricity is remaining. That high voltage of several thousand volts is being applied to the ion blower so that never touch the inside or the electrodes of this equipment.

3) High temperature hazard

The motor surface becomes hot during the operating and teaching it may result in burn injury. Thorough care must be exercised when working on the motor. A label "High Temperature" is adhered on the surfaces of this equipment that may become high temperature of 50℃ or over.

4) Pinch hazard : hands , body , etc.

There are dangerous moving parts in this equipment such as the robot, tray transfer arm, tray up/down elevator, rotary pocket, rotary head, etc. so that thorough care must be exercised.

Never operate the equipment with the safety sensor of the cover being inactivated or do not operate the equipment when the safety sensor is out of order.

5) Matters to be noted when conducting the maintenance work

During the maintenance work be sure to wear the protective gloves and boots.

6) Matters to be noted concerning the interlock

This equipment is equipped with the various types of interlock functions in order to protect the operators, products and the equipment. Never operate this equipment by making these interlock function inactivated.

7) Matters to be noted concerning covers

The equipment must be operated with all the covers being attached properly.
Do not operate the equipment with even one cover being removed.

8) Matters to be noted when pressing the start switch

Both in the auto and manual modes, be sure to confirm that there is not person or foreign objects inside the equipment and it is safe to operate before pressing each start switch.

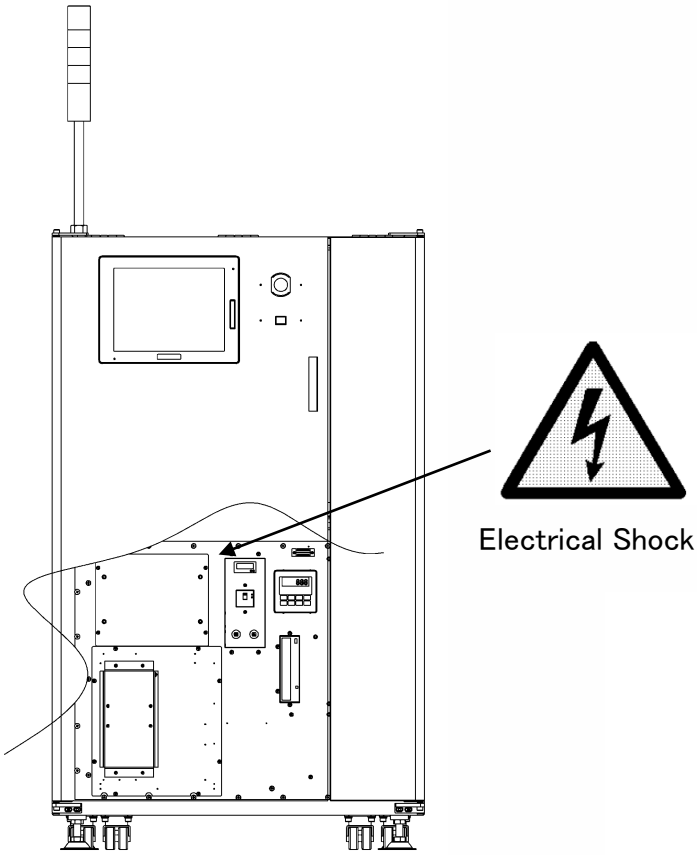
9) Inspections

Be sure to conduct the daily checkups and periodic inspections in order to confirm there is no abnormality in the equipment before conducting the operations.
In the event of contingency, confirm the possible causes and take the necessary measures for safe operations.

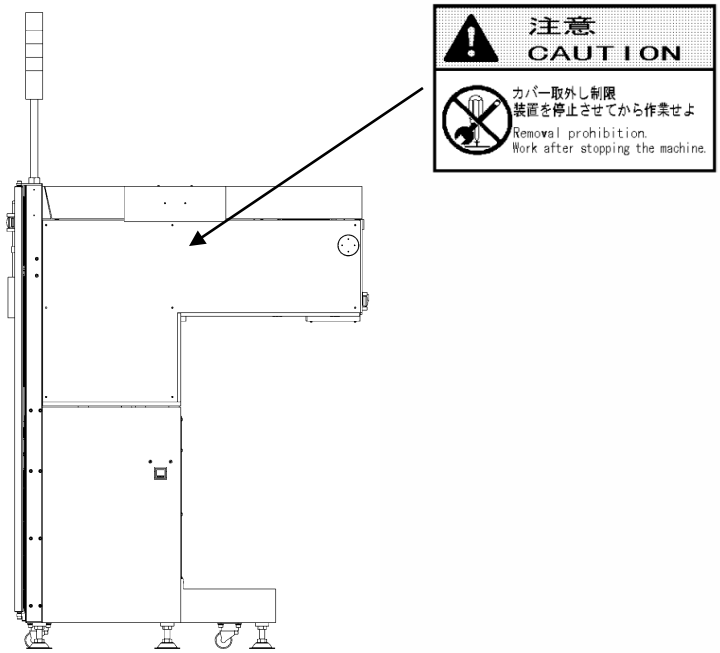
10) Risk warning labels

On the IC handler, label to warn the above content has been affixed in the vicinity of the relevant point. Enough attention give as thank you.

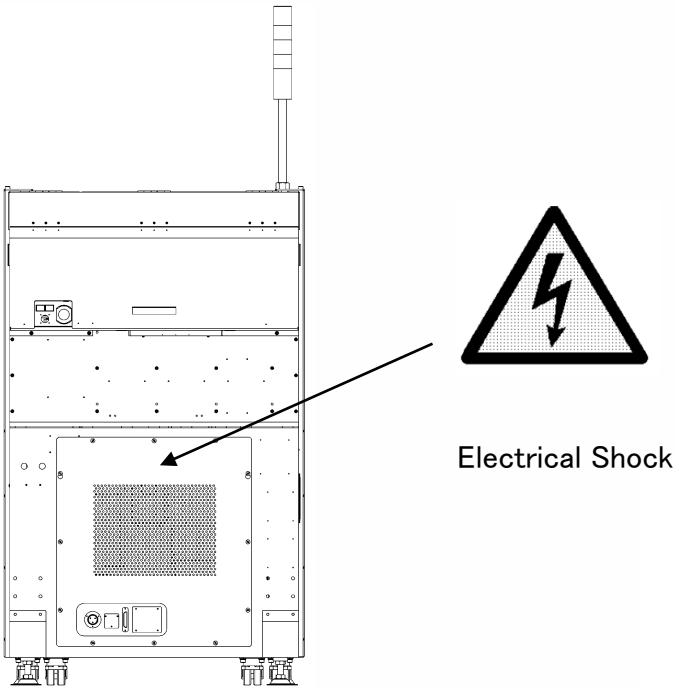
[FRONT]



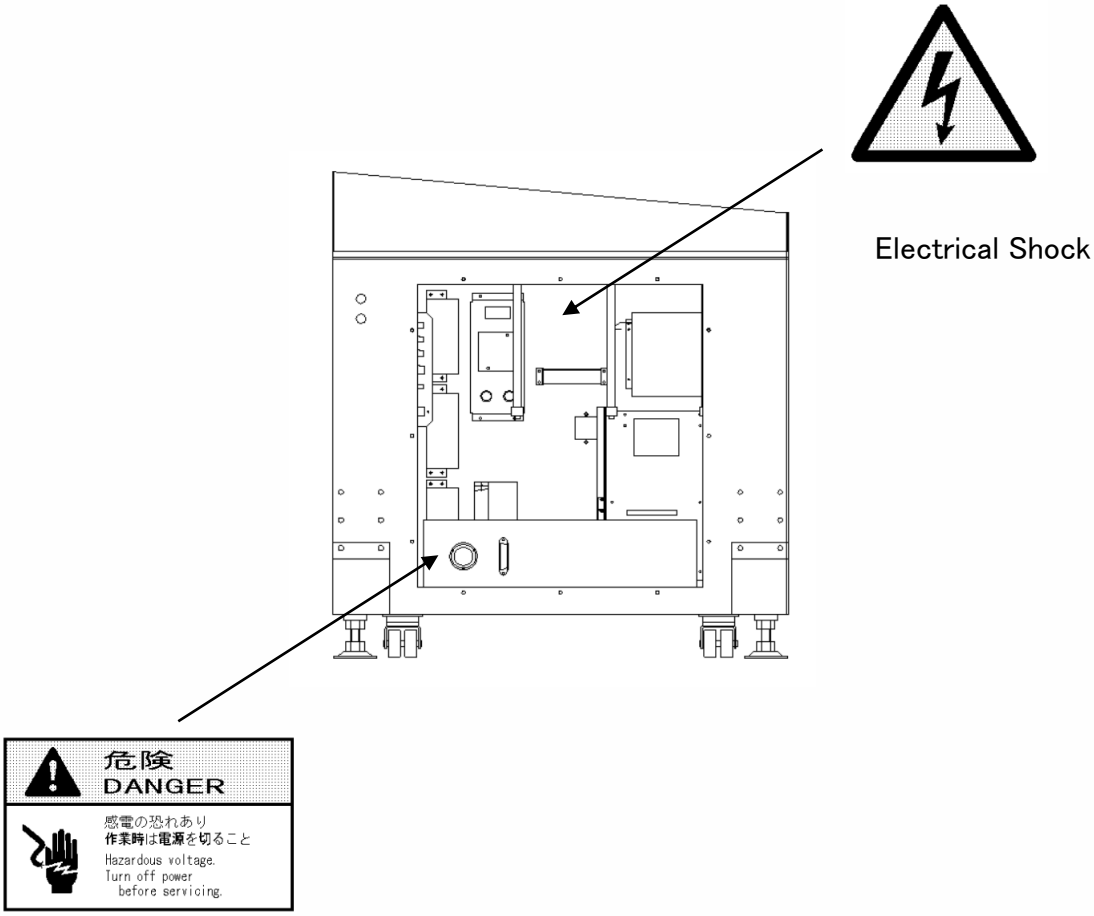
[BOTH SIDES]



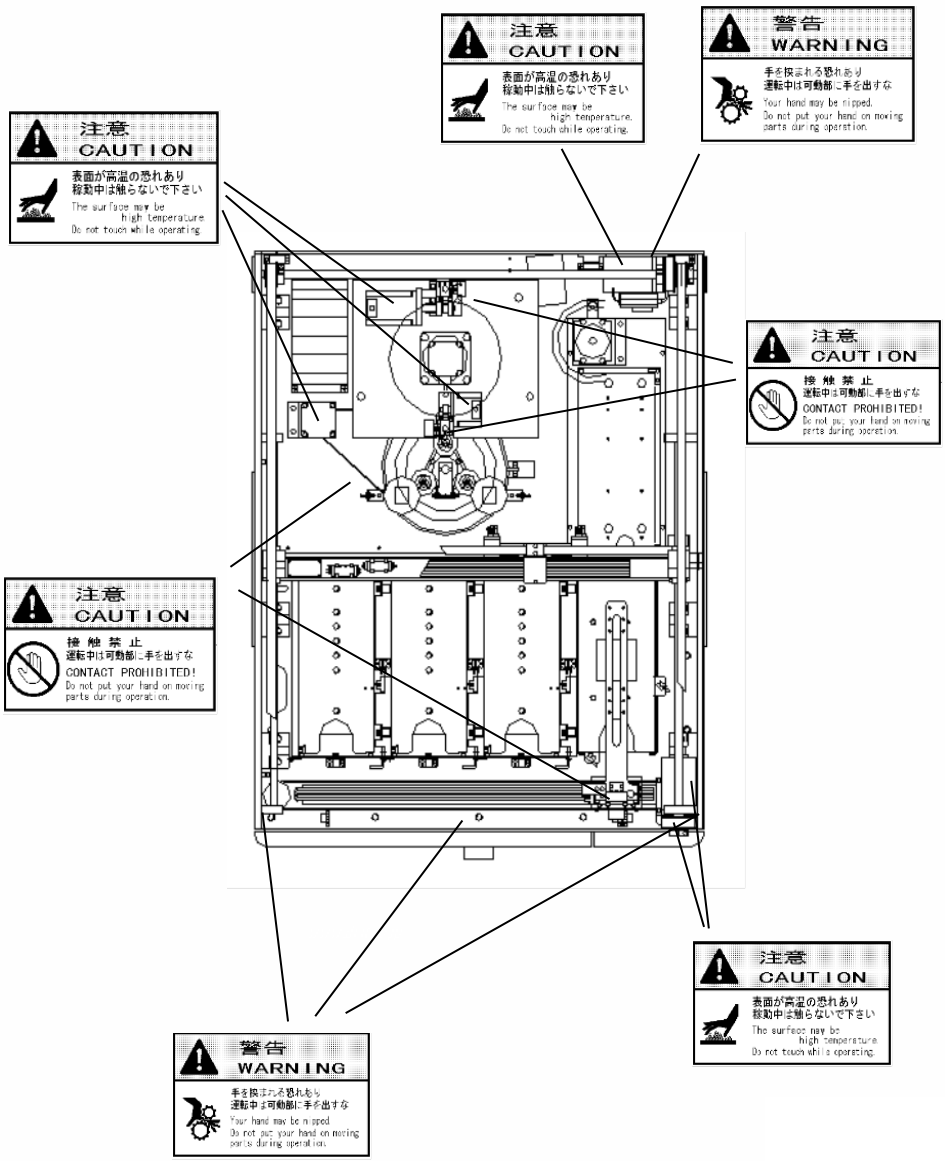
[REAR]



[CONTROL UNIT]



[MECHANICAL PARTS]



2.Outline of the equipment

2.1 Specifications

1) Name of the equipment:	Small sized high-speed handler
2) Model:	SAH-2200
3) Utilities:	AC100V 15A single-phase,50/60Hz 0.4～0.7MP,50 NL/dry air
4) Dimensions:	W 832×D 1,038(*1,088)×H 1,380 *:L-Type
5) Weight:	approx. 360 Kg
6) Equipment protection grade:	IP20 or equivalent
7) Installation environment:	Electrical equipment installation location protected against water Operation temperature:10～30℃ Operation humidity:25～85RH% (There shall be no condensation)

2.2 Installation and Handling etc.

重要
Important

1) This equipment is not waterproof, dust-proof or explosion proof type so that never install this equipment in the following locations.

a)A location where the equipment may be exposed to water or other liquid.
b)A location where the condensation occurs.
c)A location where the equipment may be exposed to the flammable gas, liquid or solid.
d)A location where the equipment may be damaged by the corrosive gas or liquid.
e)A location where dust and particles are generated.
f)A location where the equipment is exposed to the direct sun.
g)A location where the voltage fluctuation is large or there are a lot of external noises.
h)A location where the temperature changes rapidly.
i)Near the heat source.
j)A location of high temperature and high humidity.
k)A location of which floor strength and rigidity is not enough.

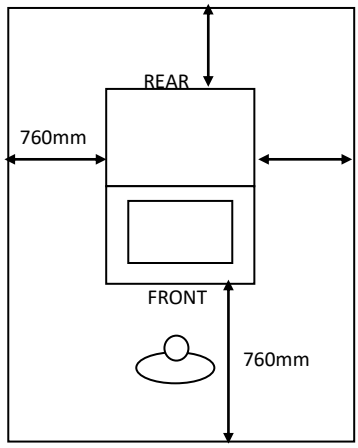
2)When unpacking, be sure to leave the equipment in the ambient temperature for a sufficient period of time in advance in order to prevent the condensation.

3)In case it is necessary to move the equipment, it must be installed on the flat surface.

4)When installing the equipment on the sloped surface, irregular surface or other instable site, the thorough care must be exercised in order to prevent overturning.

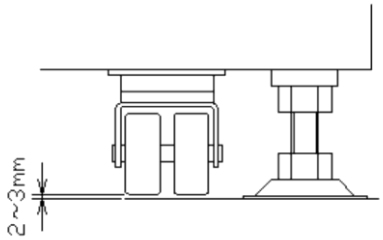
5)Unpacking and/or transportation of the equipment must be conducted by more than two persons wearing the safety shoes and protective gloves under the supervision of the supervisor.

6)For safety, secure the space for the equipment installation as per shown in the drawing below.



7)Set six adjustable feet as per shown in the drawing below. Also, be sure to fasten tightly the lock nuts after installation.

重要
Important



8)When transporting the equipment, the moving components inside the equipment must be protected from the impact during the transportation.

重要
Important

9)Also, in the case of transporting the equipment a long distance or for long hours, in order to prevent the development of rust, cover the equipment with the airtight package and put in it a sufficient amount of drying agent such as silica gel to keep the equipment dry.

重要
Important

10)In case of storing the equipment, in order to prevent the development of rust, be sure to store it in the air-conditioned room whose humidity is 60% or under and no dew condensation is expected.

重要
Important

2.3 Connection of utilities

重要
Important

1)This equipment is equipped with the breaker of 15A. Be sure to connect it to the power supply of the 15A single phase AC100V so that the operating current of the breaker can pass safely. The equipment cannot be connected to 110V or 115V line. In such case, be sure to transform the voltage to 100V using a transformer having the sufficient capacity and the short-circuit protection function at the primary side.

重要
Important

2)Be sure to use the dedicated power source for the equipment and never connect other equipment to this power supply.

重要
Important

3)Be sure to check to confirm that the power connections are tightly connected, that the power cord and plug have no abnormality in appearance, and that the power cord, plug, etc. do not generate abnormal heat when the power is turned on.

重要
Important

4)Connect the equipment to the dry air source whose pressure is 0.4～0.7 MP and the capacity is 50 NL/min.

重要
Important

5)Do not connect the wirings with wet hands, it is very dangerous.

危険
Danger

6)Incursion of water into the air system may cause for electric leakage and/or electric shock: Thorough care must be exercised.

重要
Important

7)If the water accumulates inside the regulator and has to be removed from the drain, be sure to disconnect the connectors of the power plug tester, etc. of the equipment and cover the equipment using the vinyl sheet or the equipment so that water never enters into the equipment. Then, drain the water carefully, and after having confirmed that the small droplets are completely dried, turn on the power again. This operation must be carried out by the qualified personnel who is trained in the handling of electricity and air, and can confirm the safety.

重要
Important

8)In order to prevent the electric shock, be sure to connect the earth cable. The grounding of this equipment falls under Class D(grounding resistance:100Ω or under)grounding work.

重要
Important

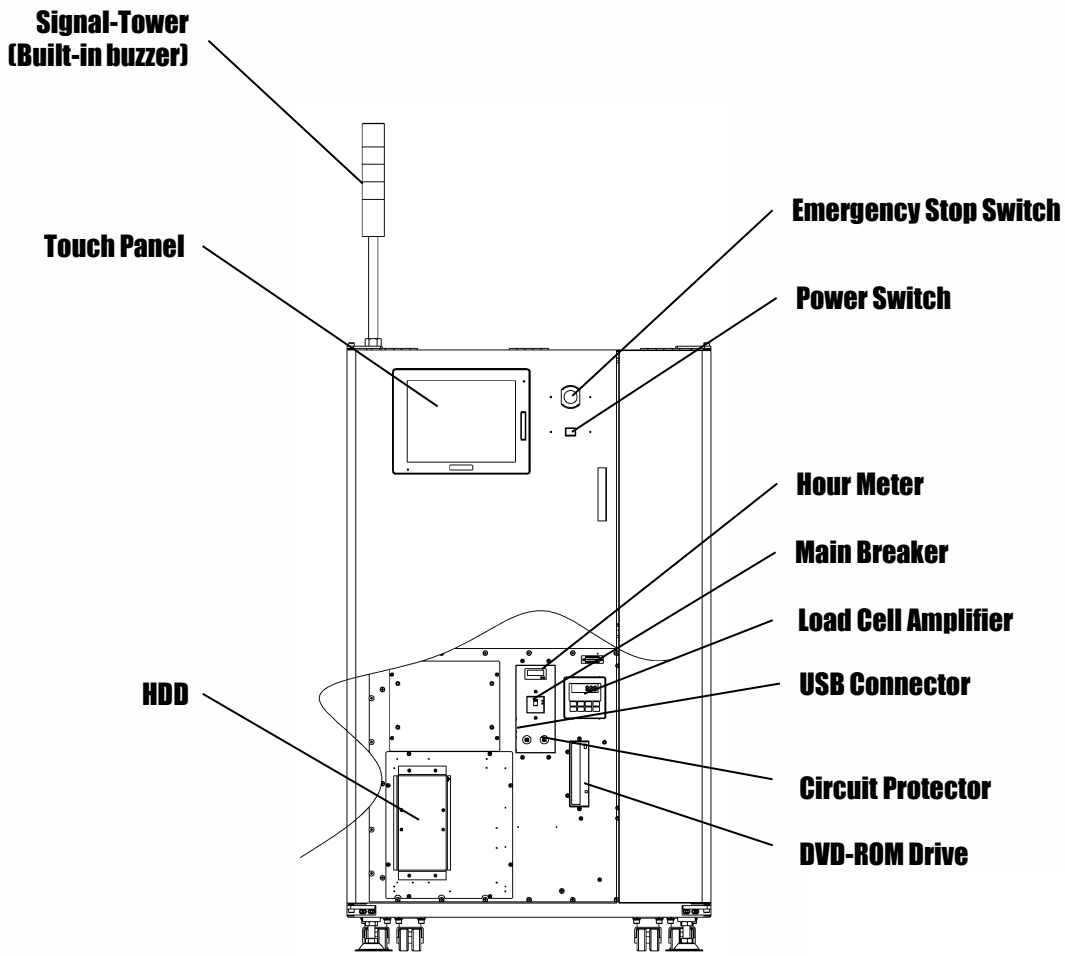
9)This equipment must be connected to the outlet having the grounding conductor.

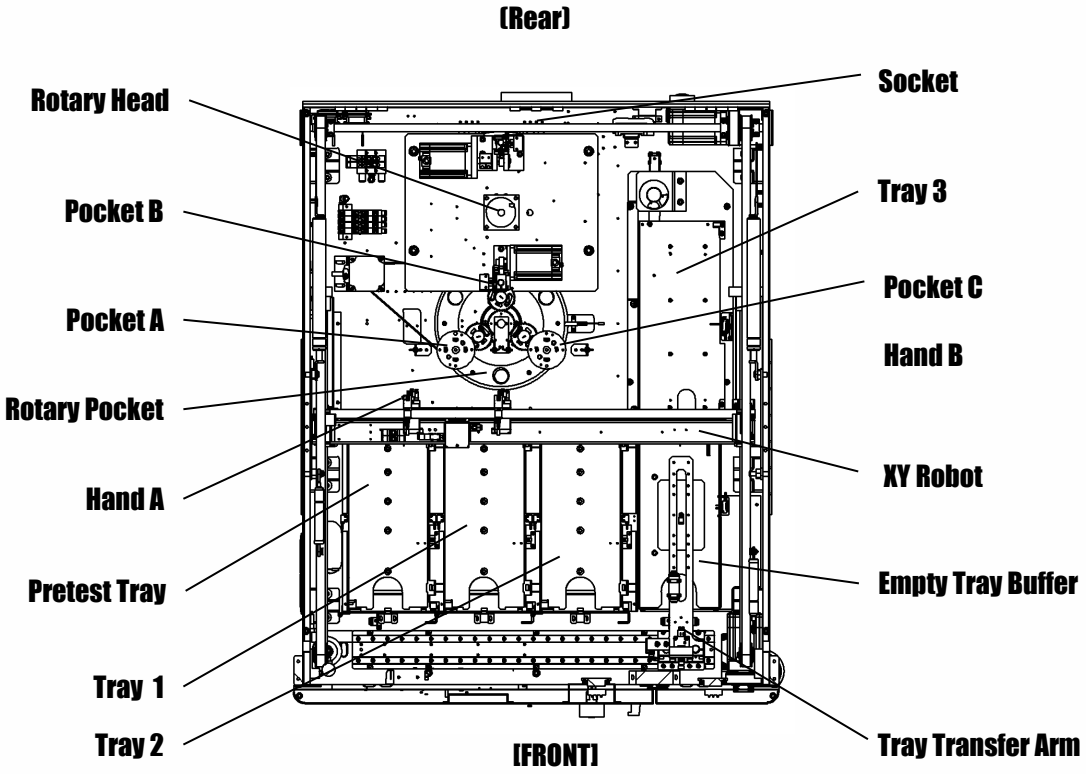
重要
Important

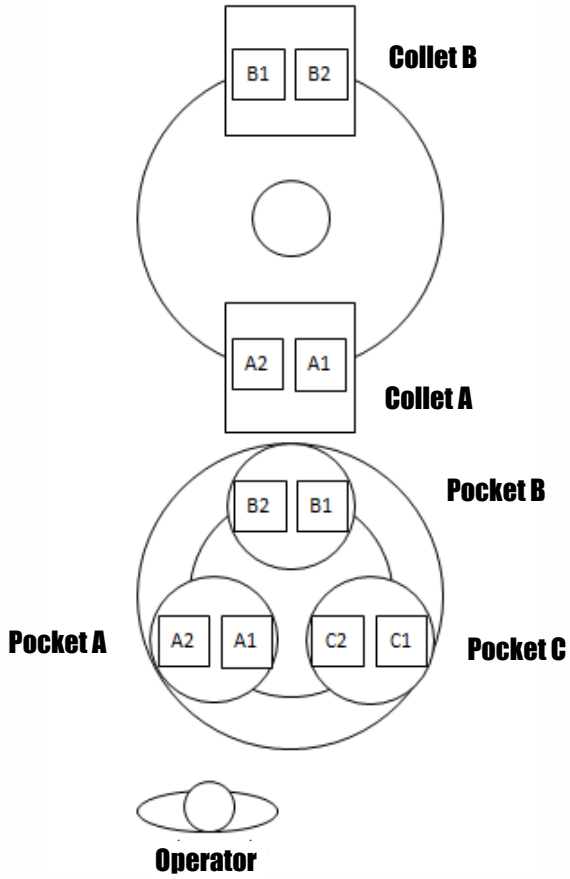
10)When making the earth connection to the equipment itself, connect the grounding cable to the terminal block with E mark or the grounding terminal marked with.

重要
Important

2.4 Name of parts




(Rear)
Rotary Head
Socket
Pocket B
Tray 3
Pocket C
Pocket A
Hand B
Rotary Pocket
XY Robot
Hand A
Pretest Tray
Empty Tray Buffer
Tray 1
Tray 2
Tray Transfer Arm
[FRONT]


B1
B2
Collet B
A2
A1
Collet A
B2
B1
Pocket B
Pocket A
A2
A1
C2
C1
Pocket C
Operator

3.Operating method

3.1 Operation preparation

3.1.1 Machine start-up

(1)Please to ON the main breaker in the front cover.

(2) Turn the power switch to ON.

(3) When the machine is started, it will rise automatically SAH-2200 applications.

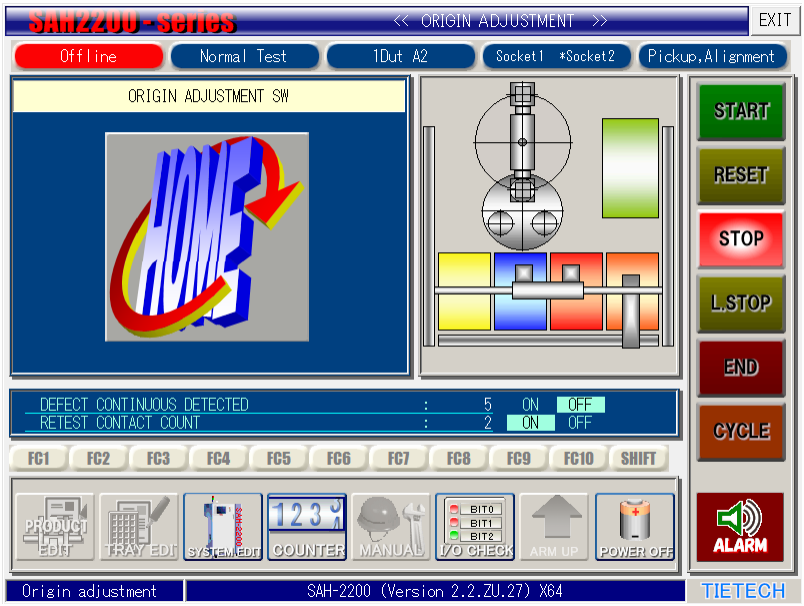


Fig 3.1A Startup screen

3.1.2 Return to origin

(1)If you press the HOME button on the startup screen to start the homing.

During homing operation, the unit of origin in the operation will be flashing in red.

Unit of origin return completion will be lit in green.

When the homing operation is completed it will be displayed product data setting screen.

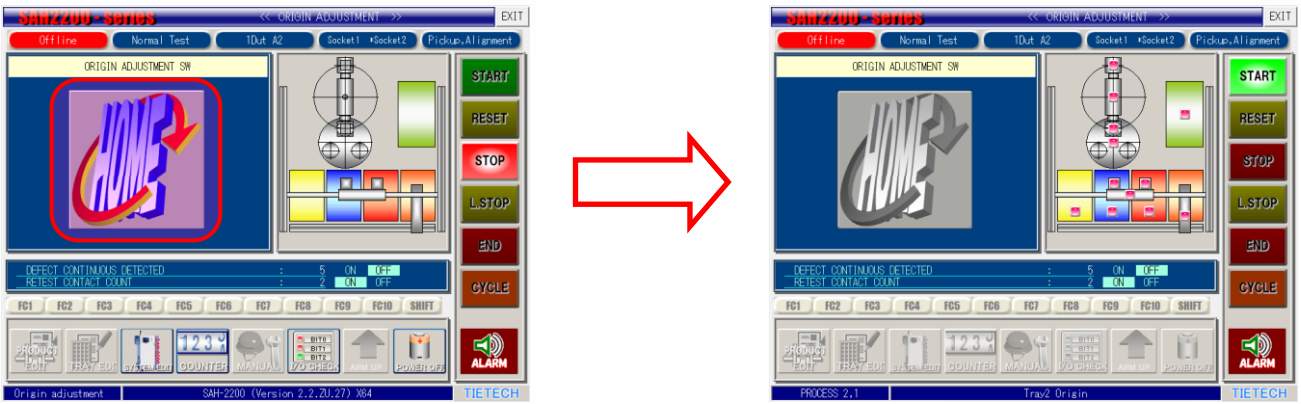


Fig 3.1B Homing in screen

3.1.3 Package data set

(1) Select a package data from No.1 which is listed among the registered data of NO.20.

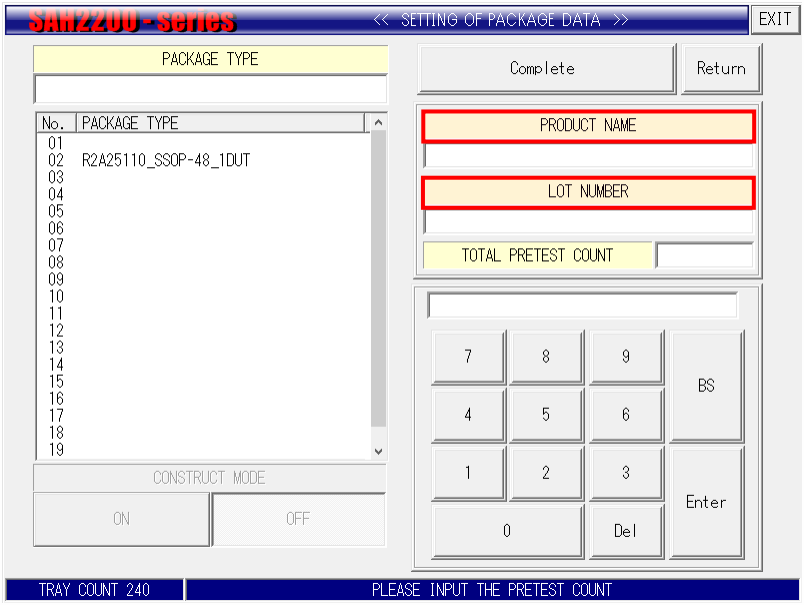


Fig 3.1C Package data setting screen

(2) PRODUCT NAME

If you want to configure the product name, press the tag of **the product name**. When the keyboard screen appears, enter the name of the product. Normally, it does not matter even without input.

Fig 3.1D Product Name input screen

(3)Lot Number

If you want to configure the lot number, press the tag of the **lot number**. When the keyboard screen appears, enter the Lot number. Normally, it does not matter even without input.

Fig 3.1F Lot number input screen

(4) Construction planned number of input

By operating the numeric keypad screen on the bottom right and enter the number of planned construction. Construction scheduled number is set by pressing the **ENTER button.**

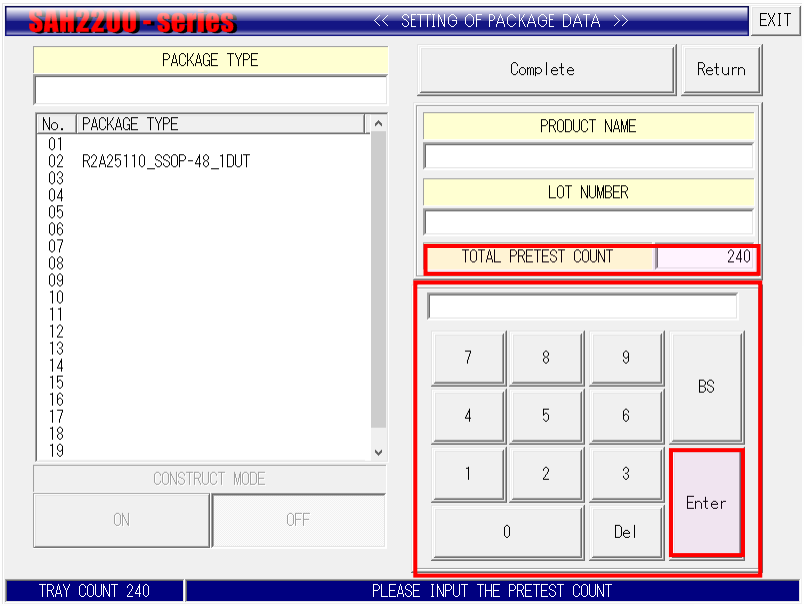


Fig 3.1G Construction planned number of input screen

The number of construction plans in case you do not want to set every time,
it will rewrite the operating environment configuration files c:¥SAH¥Data folder.
You set the Pretest key of User section of the file Config.txt.

Config.txt	Operating environment configuration file
・	
・	
[User]	Section name
pretest=9999	Default of the construction plan number
・	
・	

(5) Setting completion

When you press the setting completion button to package data set is complete.
When you press the Back button to return to the home return screen.

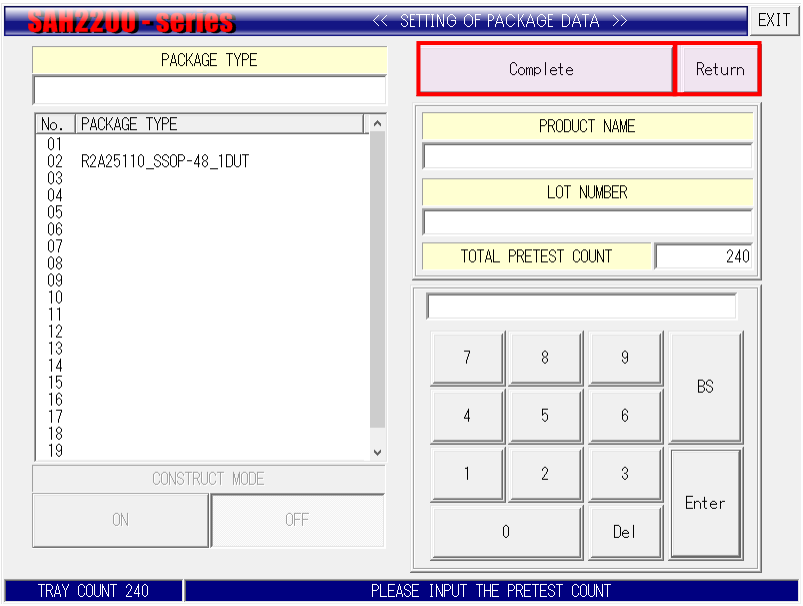


Fig 3.1H Package data setting completion screen

3.1.4 Ready for operation

(1) It opens stop screen when the package configuration is complete.

It will set the product tray to supply tray. It will set the empty tray from the tray 1 tray 3.

Fig 3.1L Stop screen

(2) Automatic operation start

Press the **start button** Once you check the status of the machine inside and interlock.

It will start the automatic operation.

3.2 Operation screen

3.2.1 Homing screen

(1) Motor Hold Off

Immediately after starting the application or is the screen of the motor hold Off. Since the excitation of the motor is turned off, there is a limit to the operation of the data setting item in the state of this screen. Run the homing operation in order to remove the restriction.

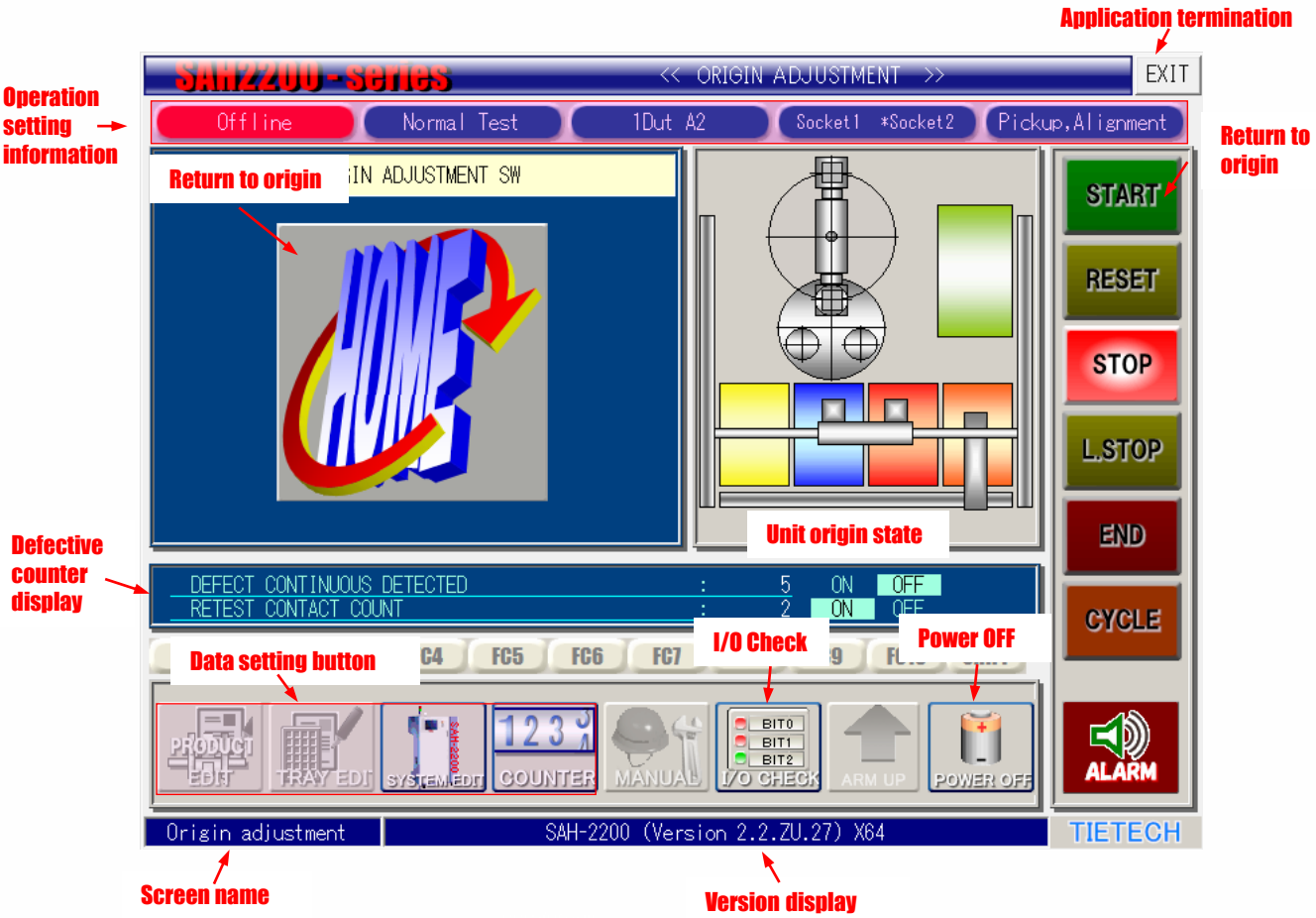


Fig 3.2A Motor Hold Off screen

(2) Motor Hold On

This is the screen of the motor hold ON state after the completion of home return. Operation of data setting item, allows manual running operation and the operation of the tray transfer arm.

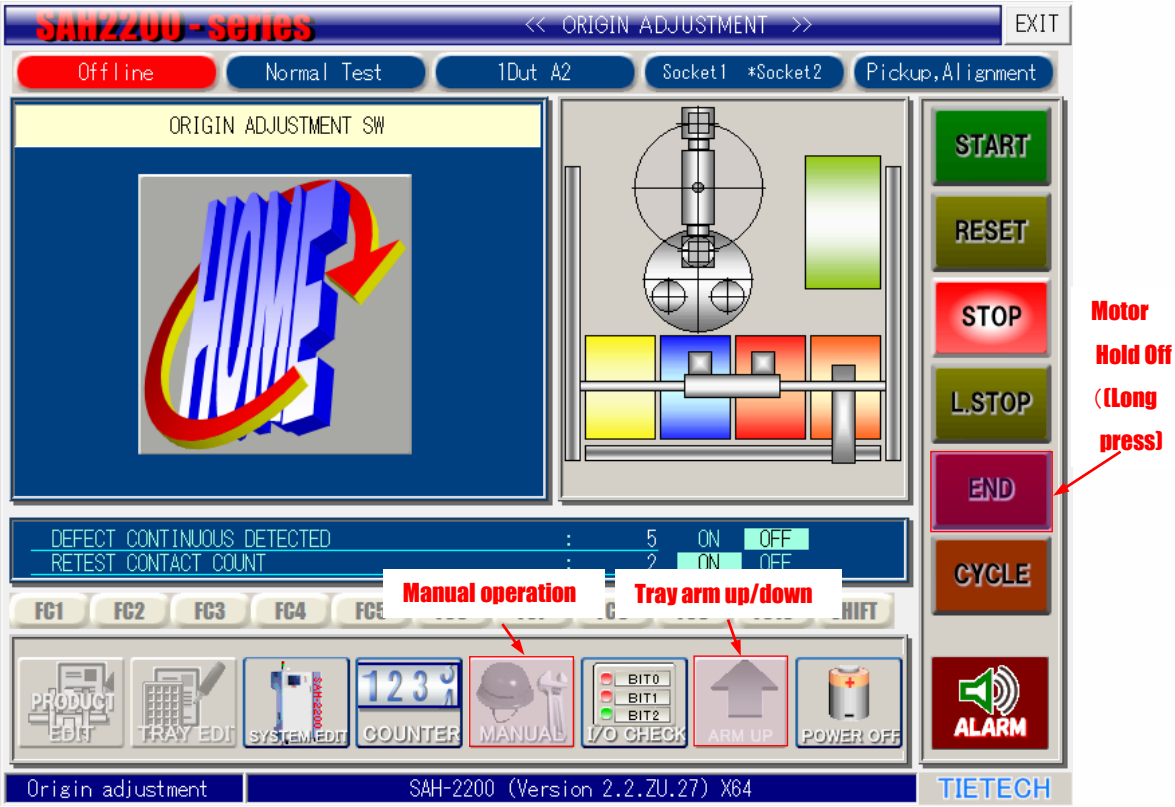


Fig 3.2B Motor Hold On screen

3.2.2 Stop Screen

This screen homing operation and package data set has been completed.

To start the automatic operation by pressing the start button.

Fig 3.2C Stop Screen

3.2.3 Automatic operation screen

It is a screen in automatic operation. If you press the stop button to stop the automatic operation.

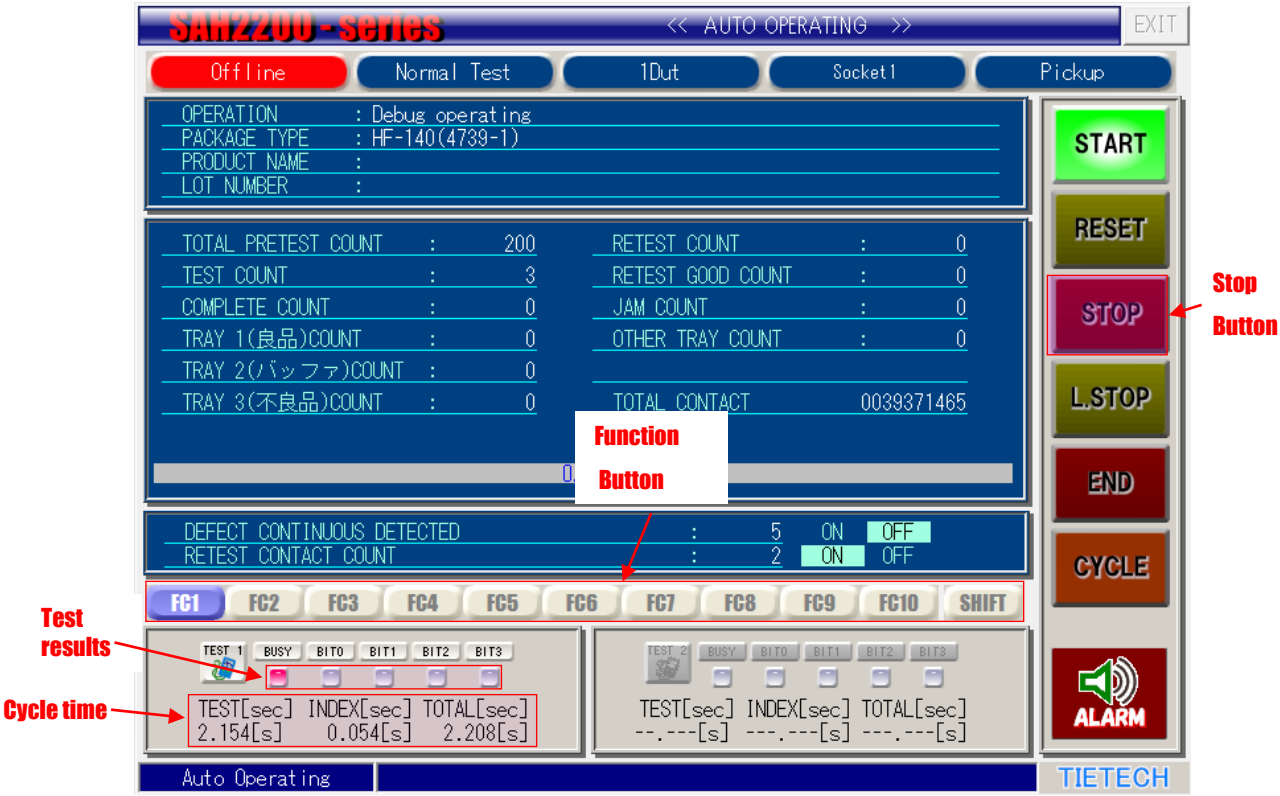


Fig 3.2D Automatic operation screen

3.2.4 Lot end screen

By automatic construction of setting count and completion or tray construction completion or 1 tray end It is a screen when the automatic operation has been completed. You can select the following operation.

(A) Go back to the origin return screen

(B) Package data selection

(C) Defective product retest

(D) Continued construction

In sustainable construction state, the start button is blinking. Press the Start button to start the continued construction.

Fig 3.2E Lot end Screen

3.2.5 Operator call screen

It is a screen at the time of operator call occurs. Depending on the operator call content, operation button is enabled. After performing the required operation, and then release the operator call by pressing the **reset button**. For details on the contents of the operator call, we'll explain in Chapter 4.

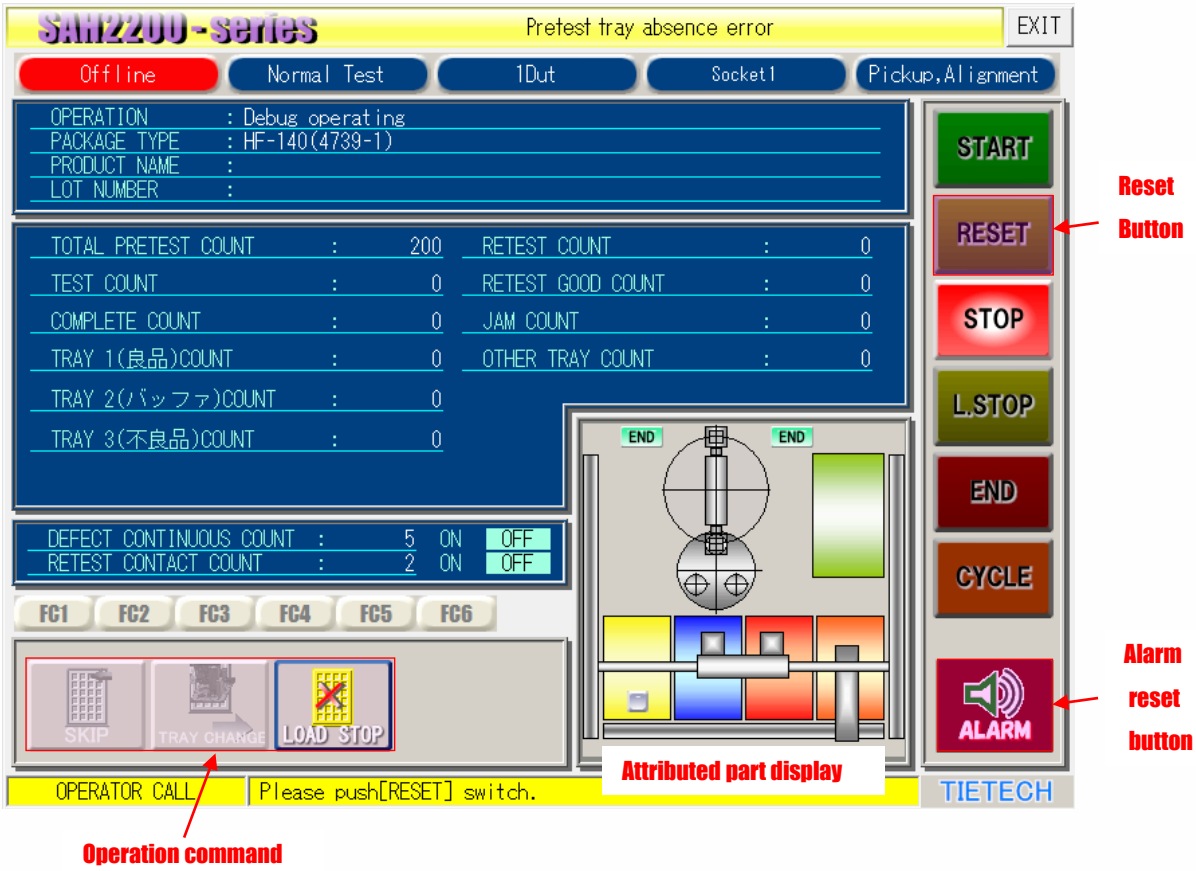


Fig 3.2F Operator call screen

3.2.6 Error stop screen

It is a screen at the time of error stop. By error content, the operation button is enabled.

After performing the required operation, press the **reset button** to clear the error stop.

The contents of the error, we'll explain in Chapter 4.

Fig 3.2G Error stop screen

3.3 Operation end

3.3.1 Application termination

(1) When you press the **EXIT button** on each screen application is terminated.

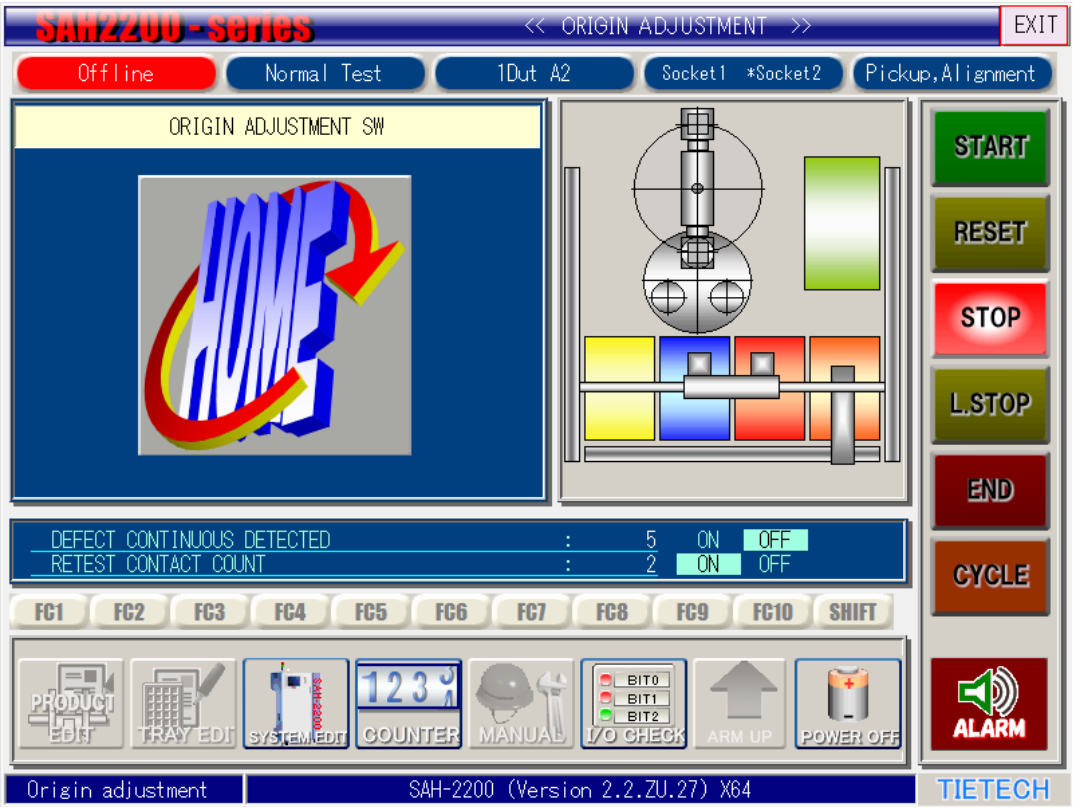


Fig 3.3A EXIT button of homing screen

(2)Exit confirmation message box appears. If you want to exit the application, press the **Yes button**. If you do not want to quit the application, press the **No button**.

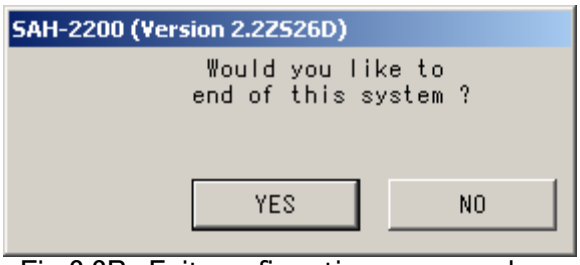


Fig 3.3B Exit confirmation message box

3.3.2 Power Off

(1) Press the power button Off of origin screen or stop screen or lot end screen.

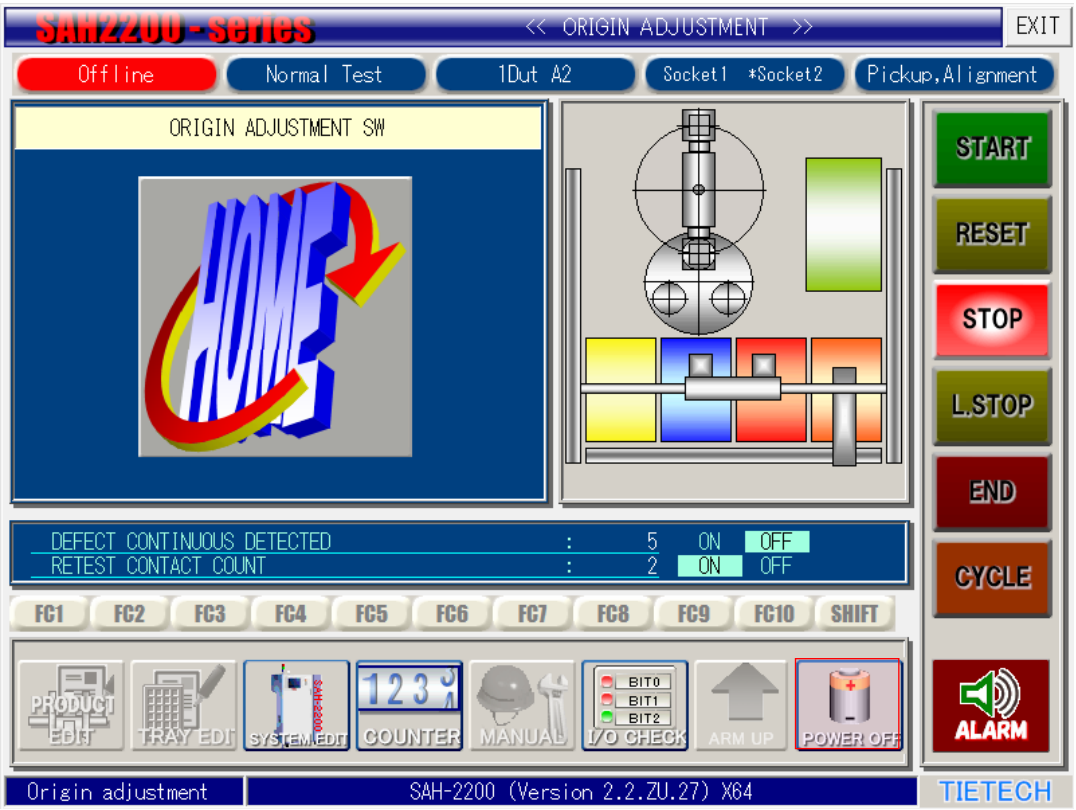


Fig 3.3C Power OFF of homing screen

(2) Power OFF confirmation message box appears. If you are a power OFF, press the Yes button. If you do not want the power OFF, press the No button.

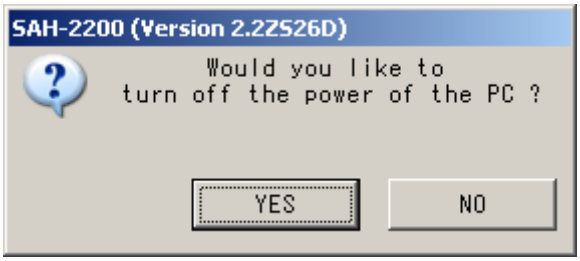


Fig 3.3D Power OFF confirmation message box

3.4 Automatic operation

3.4.1 Automatic operation

(1)自動運転画面は以下のようになっています。

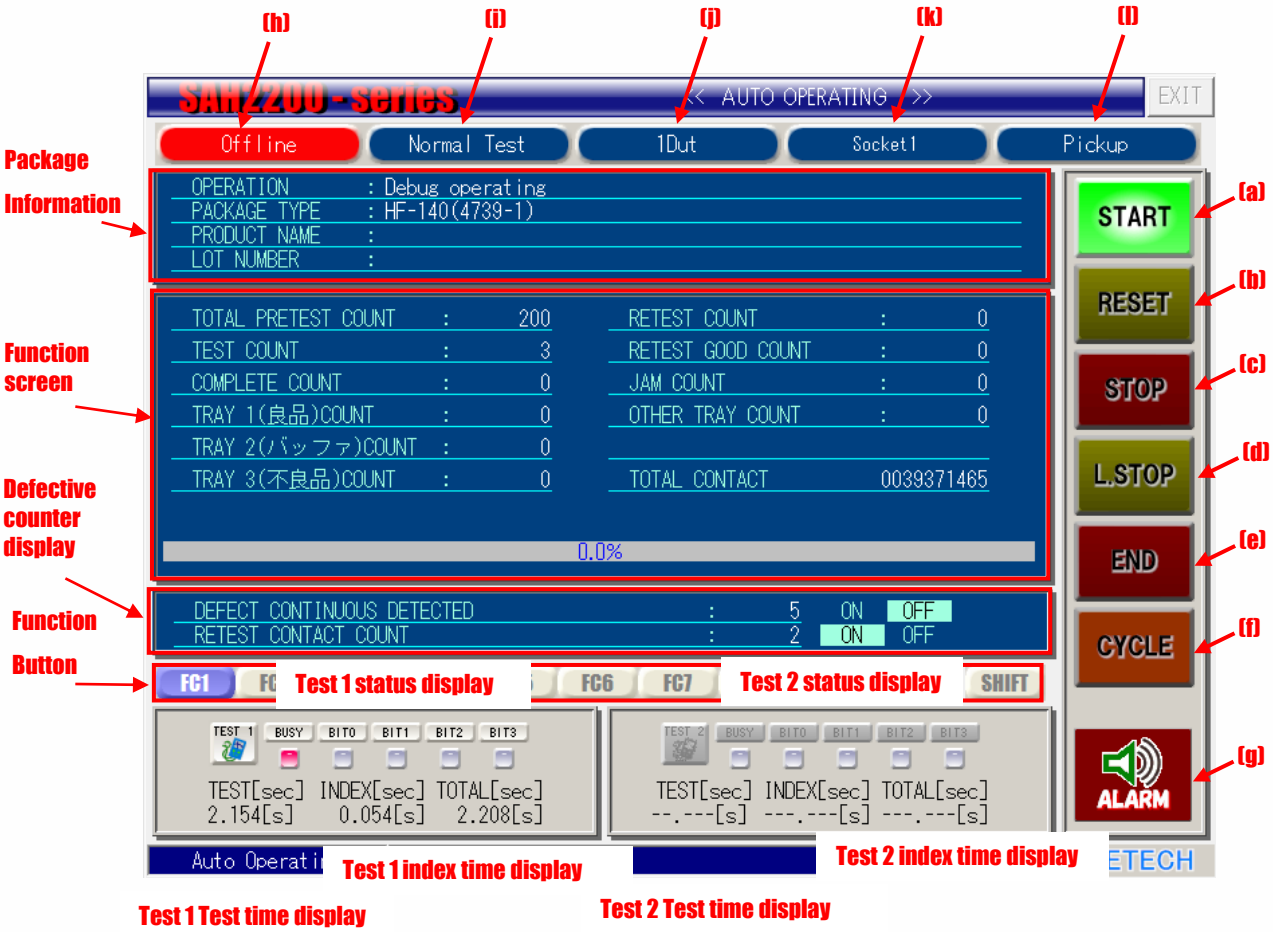


Fig 3.4A Automatic operation screen

(a)Start button

It will start the automatic operation.

(b)Reset button

Nothing.

(c)Stop button

It will stop the automatic operation.

(d)Load stop button

Stop the product supply, and exit dispensing the product in construction. Tray elevator will all down.

(e)End button

It will end after all the product starts in the current tray. Tray elevator will all down.

(f)Cycle stop button

1 cycle operation will stop. Unit to cycle stop is set by the operating environment setting screen.

(g)Alarm reset button

Alarm sound (buzzer) will OFF.

(h) Test status display

It will display the online or offline test state.

(i) Test mode display

It will display the test mode of the normal test or noise test or high temperature test or isolation test.

(j) DUT status display

It will display the status of the 1DUT or 2DUT.

(k) Use socket Display

It will display the status of the socket to be used.

(l) Socket storage mode or pressure monitor or alignment status display

Socket storage mode, you can display use state of the pressure monitor, the presence or absence of the alignment unit.

3.4.2 Function screen

(1)Function 1

Displays information about the number of products in construction.

Fig 3.4B Function 1screen

(a)TOTAL PRETEST COUNT : Construction scheduled number

(b)TEST COUNT : Number of IC handler has started construction

(c)COMPLETE COUNT : The number stored in the tray

(d)TRAY1 COUNT : Number of IC handler has stored classified into the tray 1

(e)TRAY2 COUNT : Number of IC handler has stored classified into the tray 2

(f)TRAY3 COUNT : Number of IC handler has stored classified into the tray 3

(g)RETEST COUNT : Retest number

(h)RETEST GOOD COUNT : Retest good number

(i)JAM COUNT : Jam product storage number

(j)OTHER TRAY COUNT : Other tray (tray 4 to tray 8) Storage Number

(k)VIBRATOR COUNT : Pocket vibration number

(l)Total Contact : Total socket contact counter

(2)Function 2

It will breakdown display the complete number and retest the number in a pie chart.

Pie chart of the breakdown is as follows.

Green	Tray 1
Red	Tray 2
Yellow	Tray 3
Orange	Other Tray

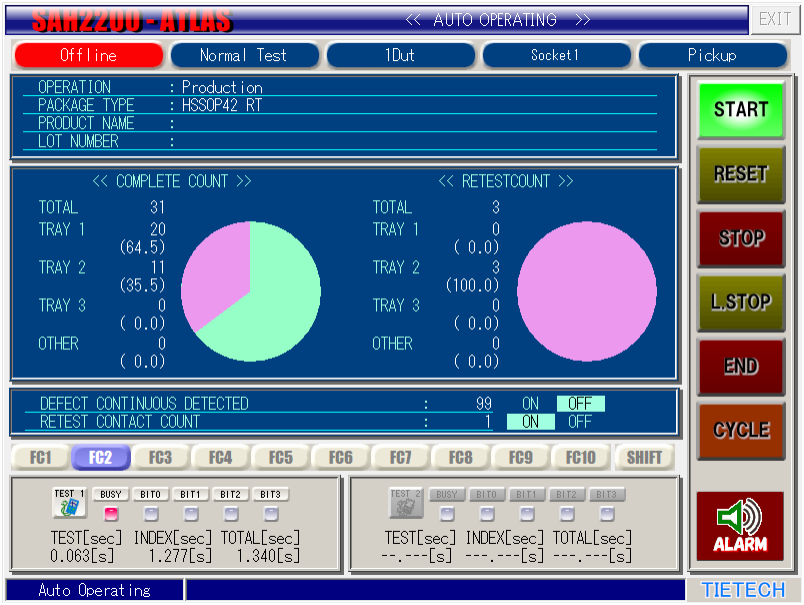


Fig 3.4C Function 2 screen

(3)Function 3

It displays the pie chart a breakdown of the number of completed by the collet.

Pie chart of the breakdown is the same as the function 2.

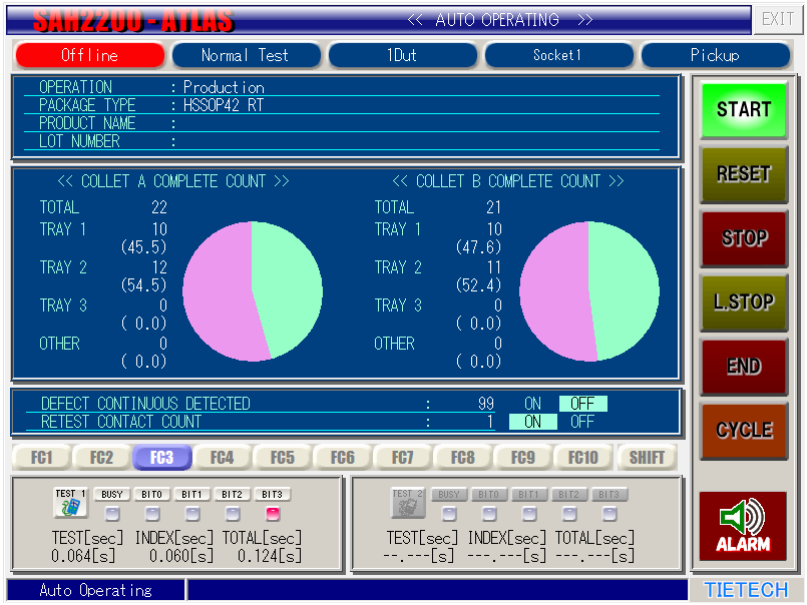


Fig 3.4D Function 3 screen

(4)Function 4

It will show a breakdown of the retest number of pie chart for each collet. Pie chart of the breakdown is the same as the function 2.

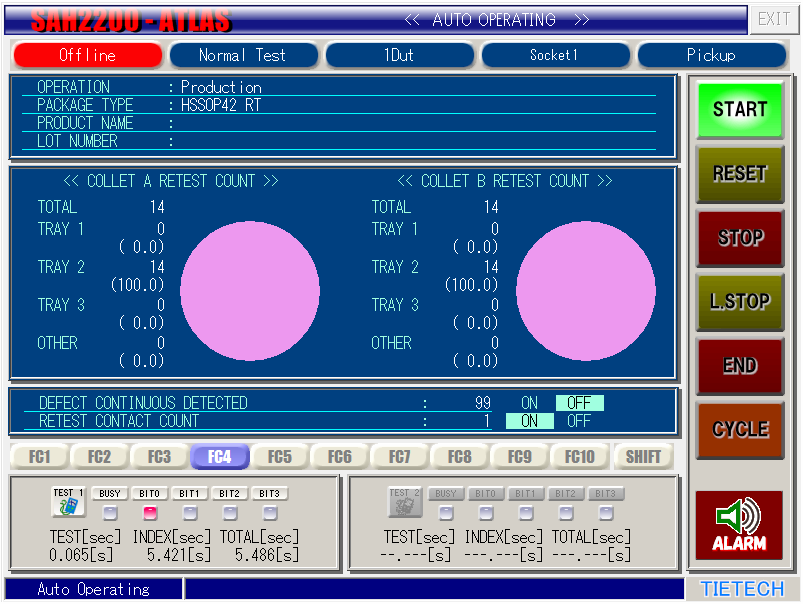


Fig 3.4E Function 4 screen

(5)Function 5

It will show a breakdown of the complete number in the BIN number of separate band graph.

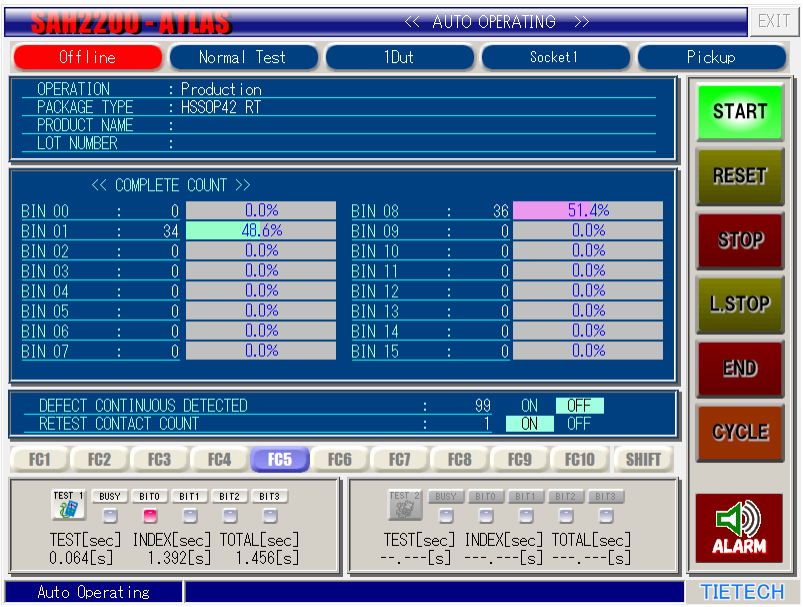


Fig 3.4F Function 5 screen

(6)Function 6

A breakdown of the complete number of collet A1, B1 collet it will appear in separate pie chart. Pie chart of the breakdown is the same as the function 2.

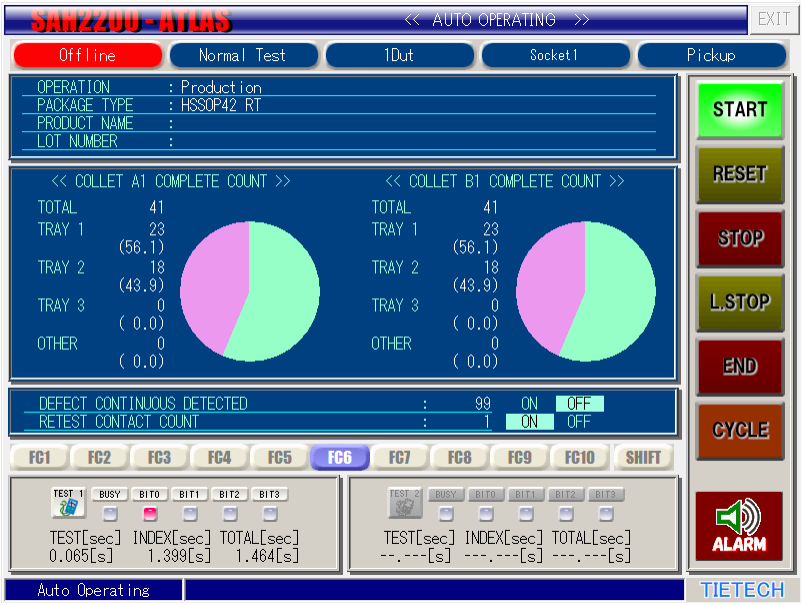


Fig 3.4G Function 6 screen

(7)Function 7

A breakdown of the complete number of collet A2, B2 collet it will appear in separate pie chart. Pie chart of the breakdown is the same as the function 2.

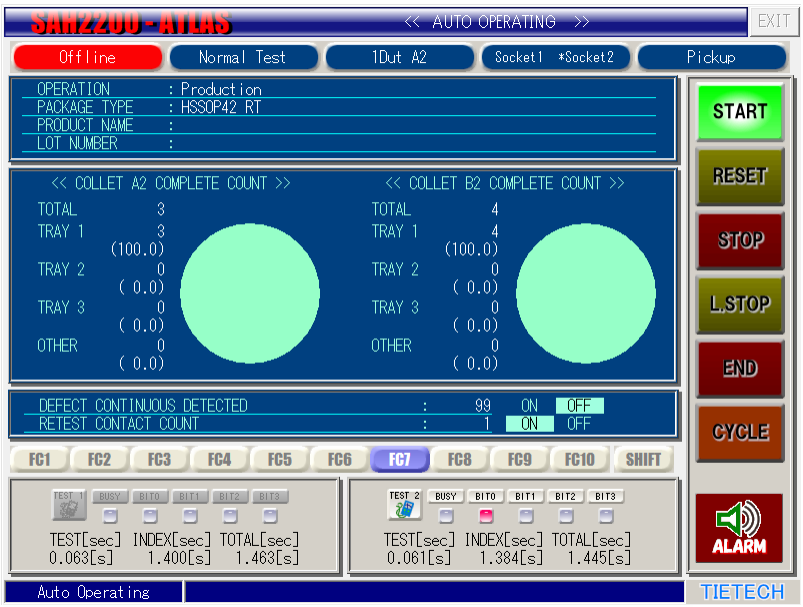


Fig 3.4H Function 7 screen

(8)Function 8

A breakdown of the retest number of collet A1, B1 collet it will appear in separate pie chart. Pie chart of the breakdown is the same as the function 2.

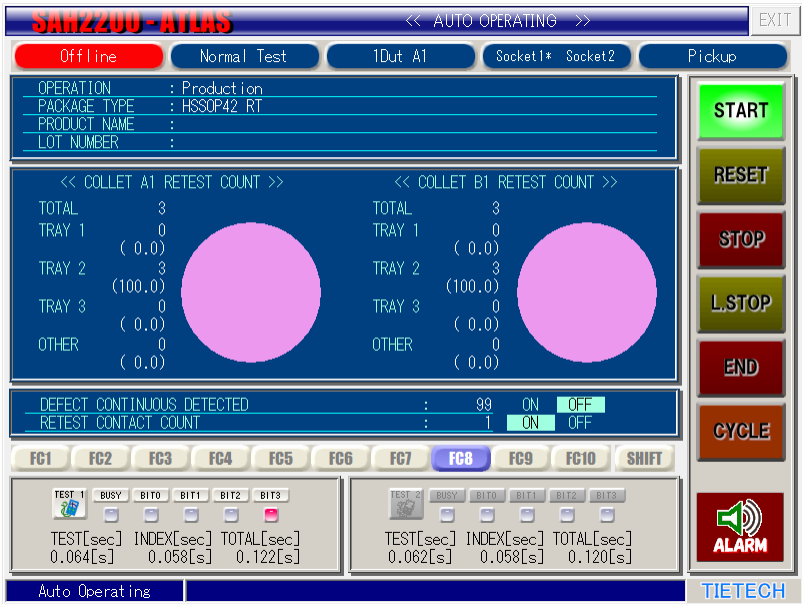


Fig 3.4I Function 8 screen

(9)Function 9

A breakdown of the retest number of collet A2, B2 collet it will appear in separate pie chart. Pie chart of the breakdown is the same as the function 2.

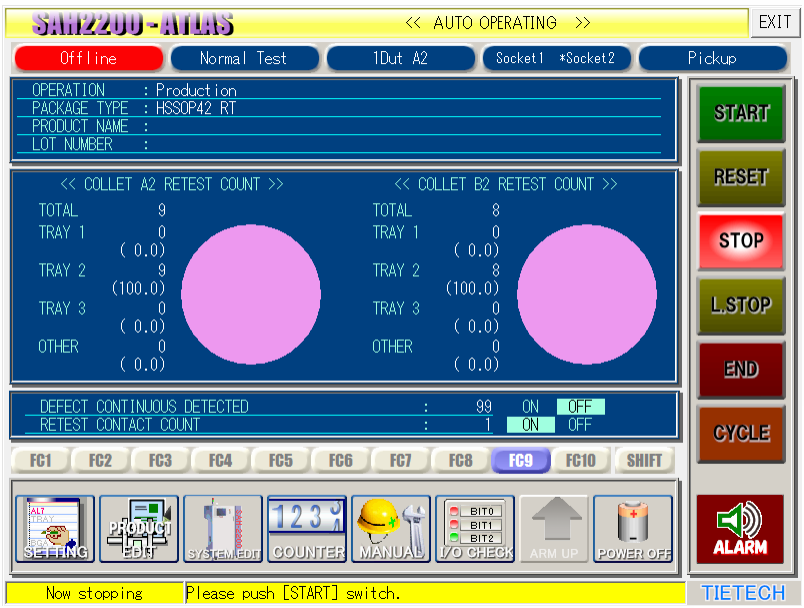


Fig 3.4J Function 9 screen

(10)Function 10

A breakdown of the complete number of socket 1, socket 2 it will show separately on the pie chart.

Pie chart of the breakdown is the same as the function 2.

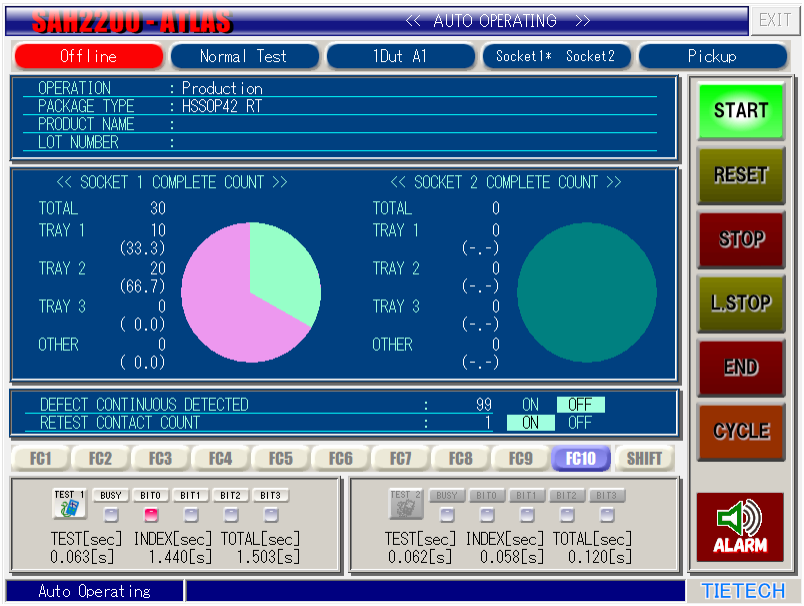


Fig 3.4K Function 10 screen

(11)Function 11

A breakdown of the retest number of socket 1, socket 2 it will show separately on the pie chart.

Pie chart of the breakdown is the same as the function 2.

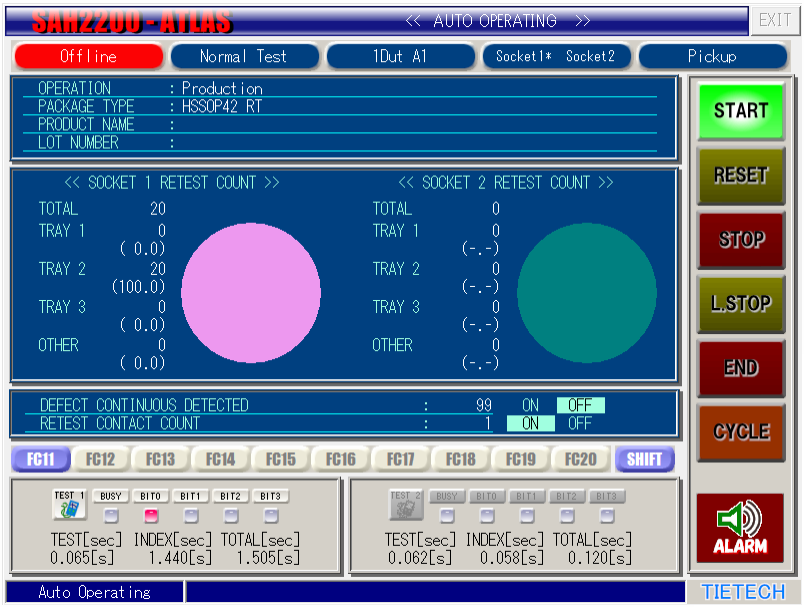


Fig 3.4L Function 11 screen

(12)Function 12

The pressure monitor measurements of the collet A it will display graph.

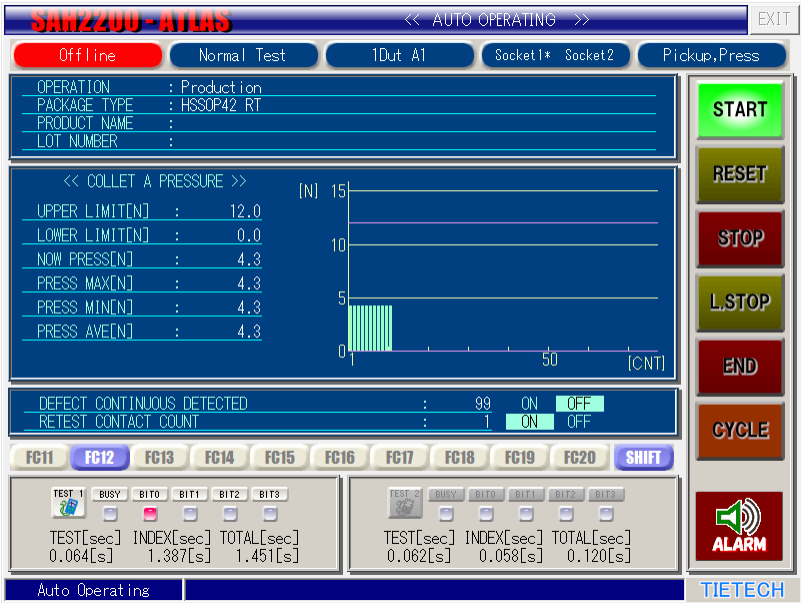


Fig 3.4M Function 12 screen

(13)Function 13

The pressure monitor measurements of the collet B it will display graph.

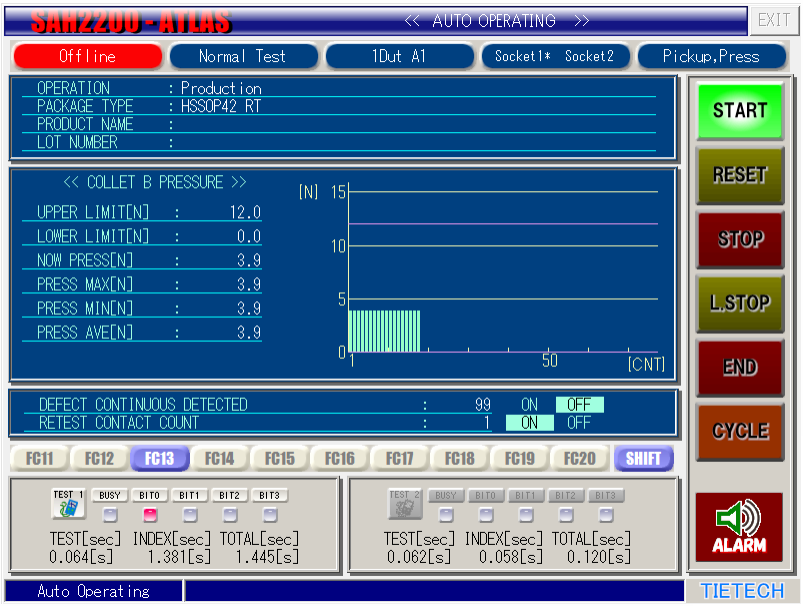


Fig 3.4N Function 13 screen

(14)Function 14

It will display the operating information of the operating time and stop time and error stop time.

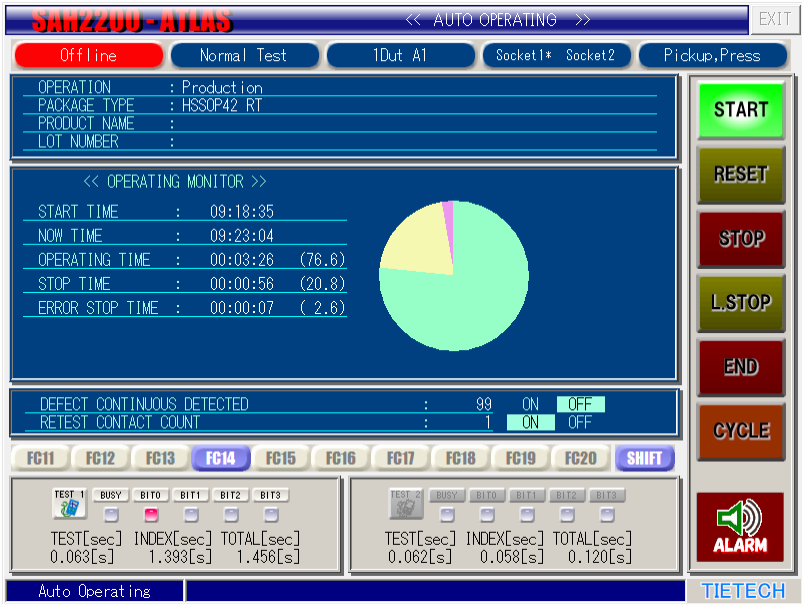


Fig 3.4O Function 14 screen

(15)Function 15

it will display the package alignment results. if there is an alignment error, error contents are displayed. The red frame shows the amount of alignment correction. The yellow frame indicates the amount of motor movement that is the amount of alignment compensation plus the collet offset and socket offset.

Fig 3.4P Function 15 screen

(16)Function 16

It will display the results of the socket yield management options.

Fig 3.4Q Function 16 screen

(17)Function 17

It will display the complete number of the breakdown of the collet A in the BIN number of separate band graph.

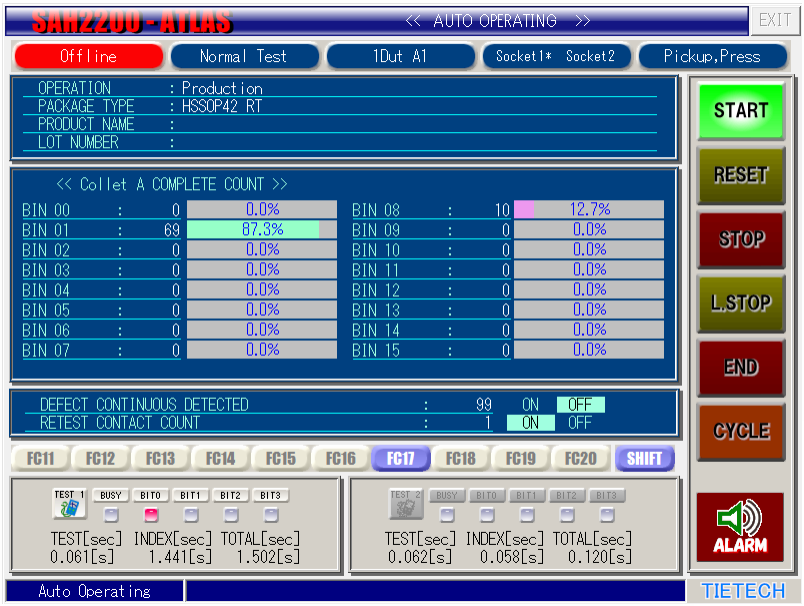


Fig 3.4R Function 17 screen

(18)Function 18

It will display the complete number of the breakdown of the collet A in the BIN number of separate band graph.

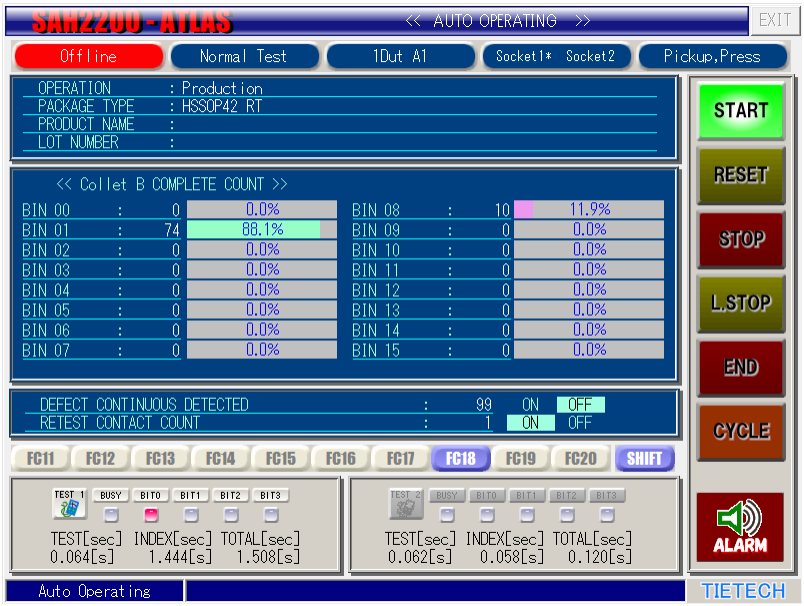


Fig 3.4S Function 18 screen

(19)Function 19

It will display information such as process information and motor position of the machine. It uses at the time of machine debugging.

Fig 3.4T Function 19 screen

(20)Function 20

It will display the error that occurred during the lot production. At the lot the end, it will be clear.

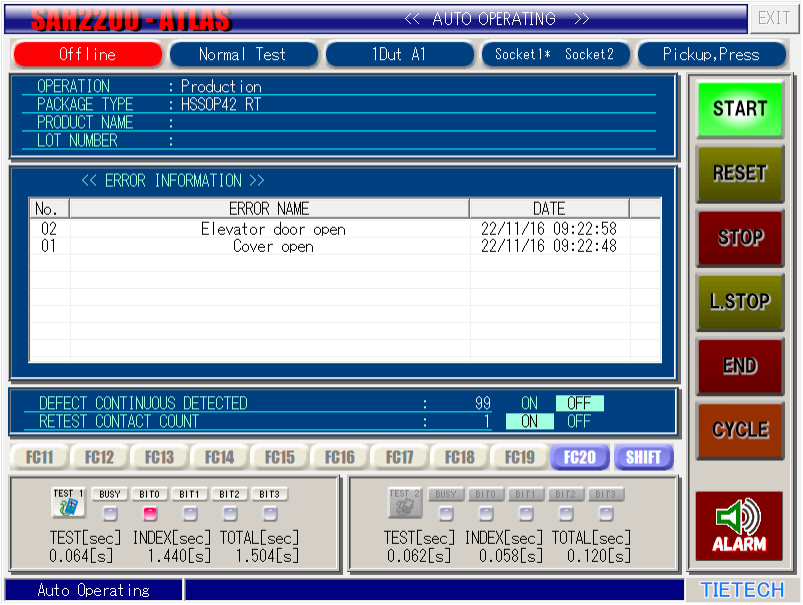


Fig 3.4U Function 20 screen

3.5 Password

3.5.1 Password input screen

If you try to run a variety of data settings, it will appear Password input screen. In order to become an important configuration changes related to equipment operation, the lock by the password is provided.

Password level is three types of supervisor, engineer and technician level.

Screen will vary depending on password level.

Fig 3.5A　Password input screen

3.5.2 Password type

Password input operation is required for the following items.

(a)All the operations of editing product data

(b)All operations of tray data editing

(c)System settings except for the counter configuration and error information setting

(d)Full operation of the manual operation

(e)I / O check screen operation

Password is provided to each supervisor and engineers and technicians, it is possible to be managed separately. Operable items are items that have been indicated as "OK" in the table below.

Item		supervisor	engineers	technicians
product data editing		OK	OK	OK
Tray data editing		OK	OK	*
	Operating environment setting	OK	OK	OK
	Motor parameters Configuration	OK	*	*
	Error information setting	–	–	–
	Tester interface settings	OK	OK	*
	Origin information setting	OK	OK	*
	Pressure monitor settings	OK	OK	*
	Counter setting	–	–	–
	Position information setting	OK	OK	*
	Timer setting	OK	OK	*
	Signal tower configuration	OK	OK	*
	Camera settings	OK	OK	*
Manual operation		OK	OK	OK
I/O check		OK	OK	OK

3.5.3 Password change

If you want to change the password, press the **key button** on the password input screen.

If you want to change the password, select the level from among the supervisors and engineers and technicians. After you enter the new password and again the new password for confirmation, allow you to change the password by pressing the **ENTER button**.

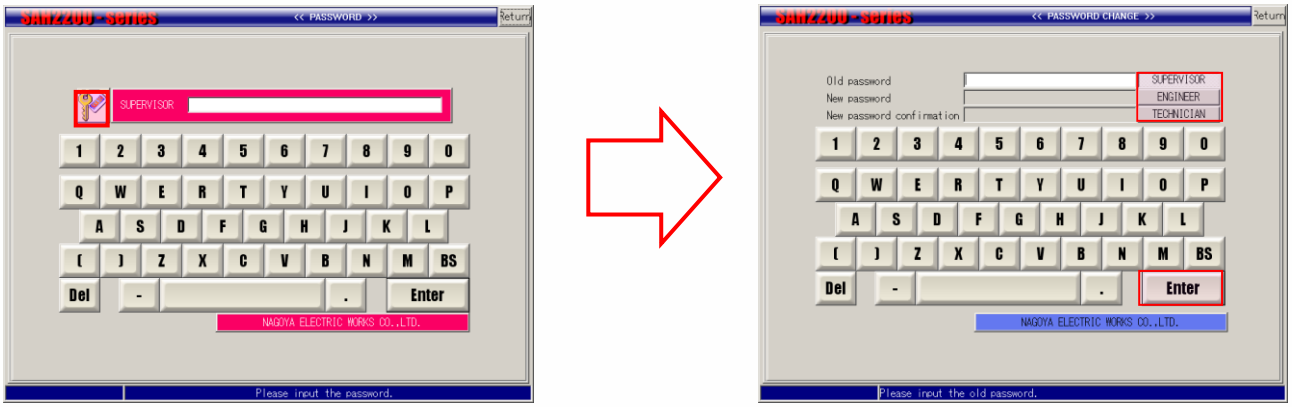


Fig 3.5B Password change screen

3.6 Manual operation screen

3.6.1 Open the manual operation screen

Press the **manual operation button** from homing screen or stop screen after completion of home return.

Manual operation menu screen is displayed.

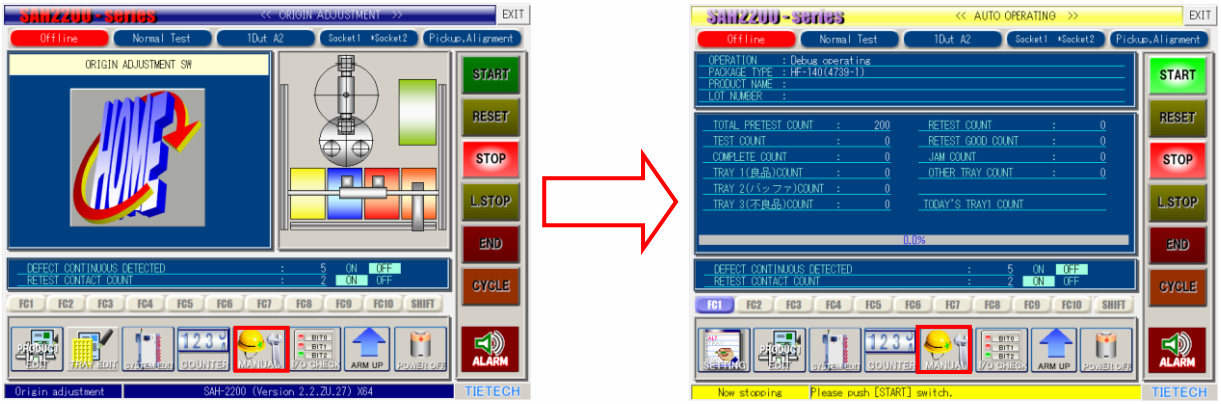


Fig 3.6A Manual operation button

Fig 3.6B Manual operation menu screen

3.6.2 I/O check

(1)I/O input

It will show the input state of the I / O. Press the input button. It will select the port number you want to check the input (port 1 to port 5).

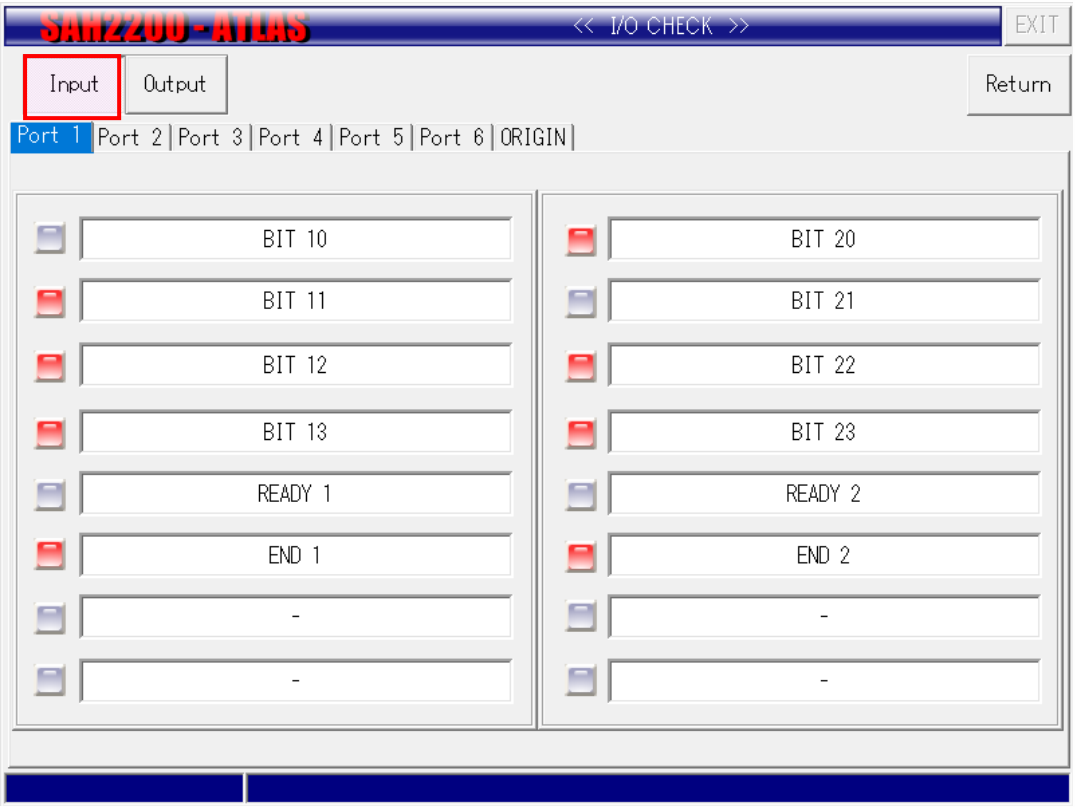


Fig 3.6C I/O input screen

(2)I/O output

It will display the output state of the I / O. Press the output button. It will select the port number you want to check the output (port 1 to port 4).

If you want to change the output, ON or OFF the appropriate bit. Homing operation will be necessary if you have changed a bit of power ON and motor hold.

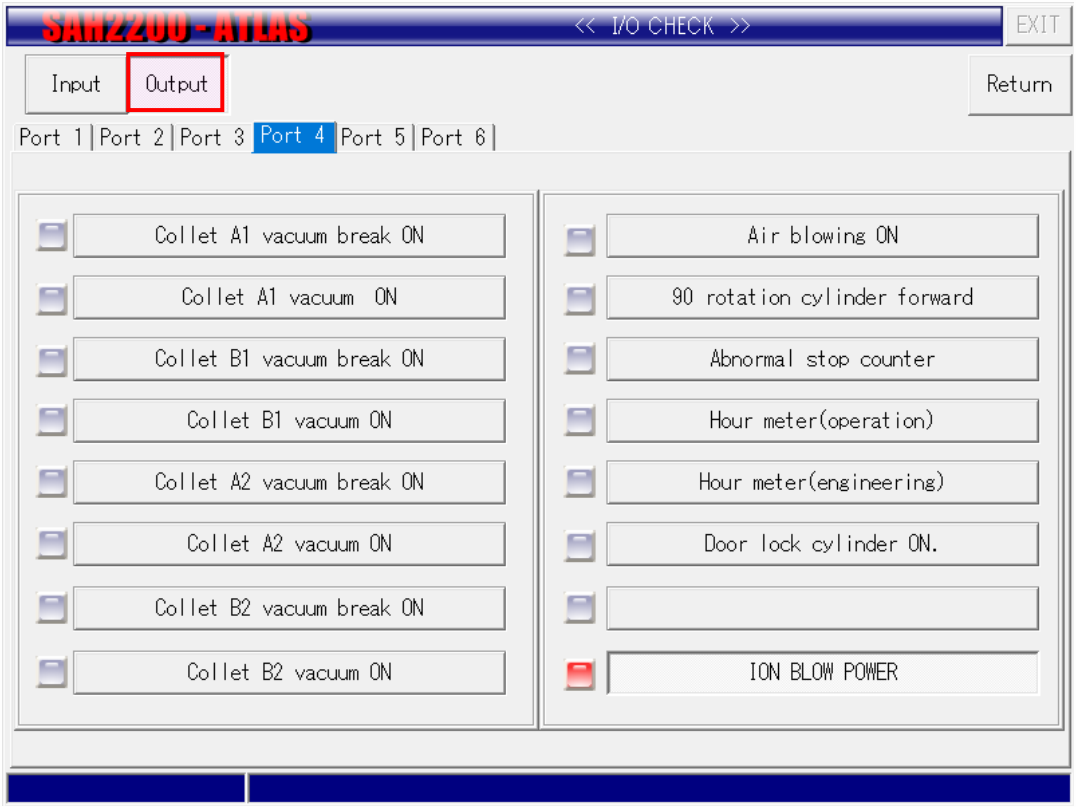


Fig 3.6D I/O Output screen

3.6.3 Manual elevator operation

(1) Elevator operation

It will operate the PRETEST TRAY elevator and TRAY 1 elevator and TRAY 2 elevator individually.

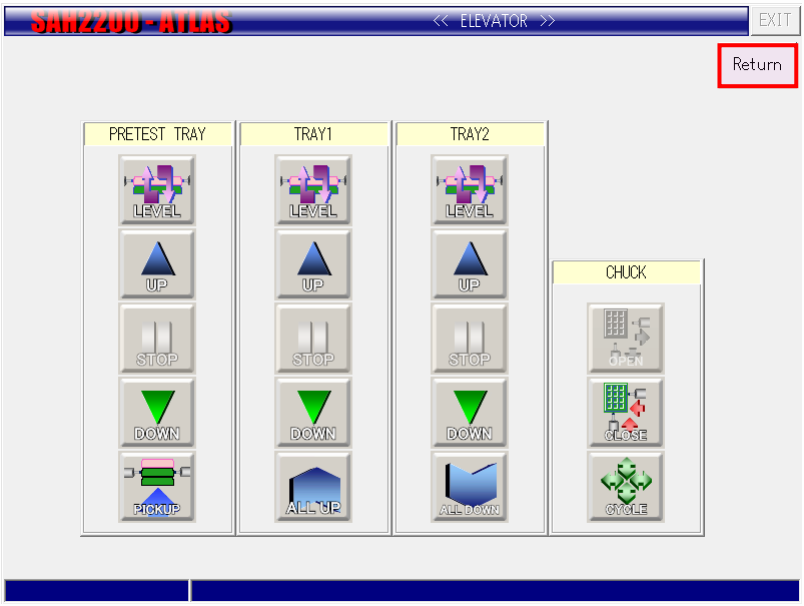


Fig 3.6E Manual elevator screen

■Level button: It makes the leveling of the elevator.

■Up button: It will move to above the elevator.

■Stop button: The elevator in operation it will be stopped.

■Down button: It will move the elevator down.

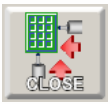
■Pickup button: Elevator rises to be able to replace the first tray, and then clamp the second tray.

2) Chuck operation

It operates the PRETEST TRAY and the TRAY 1 and TRAY 2 chuck.

■Chuck open button: It will open the elevator of the chuck.

■Chuck close button: It will close the elevator of the chuck.



■Chuck cycle button: It makes the opening and closing operation of the elevator chuck.

3)Manual menu screen is displayed when you press the **Return button**. During automatic operation, if you do not return the state of the chuck to the original, **return button** it does not take effect.

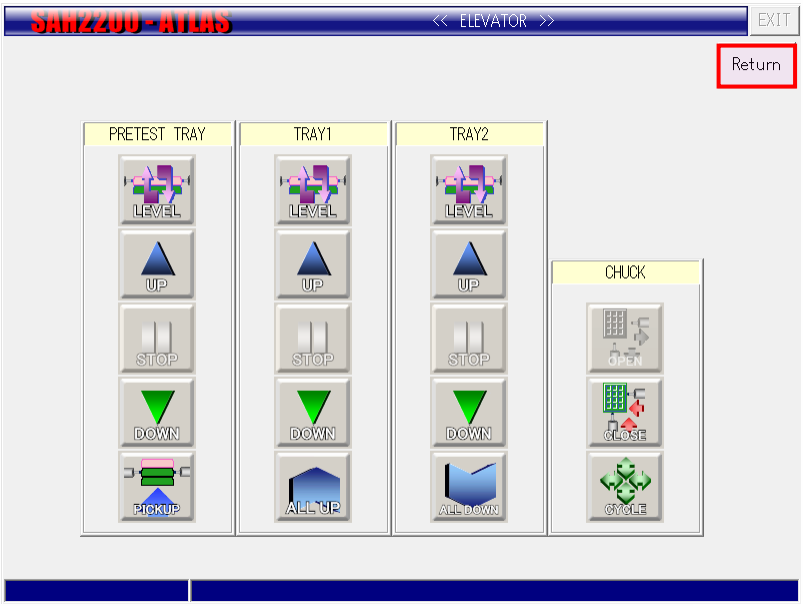


Fig 3.6E1 Manual elevator screen

3.6.4 Manual tray transport

(1)Robot movement position

Tray transport arm will be moved to the elevator position. If the tray transport arm is raised, tray transport arm is movable.

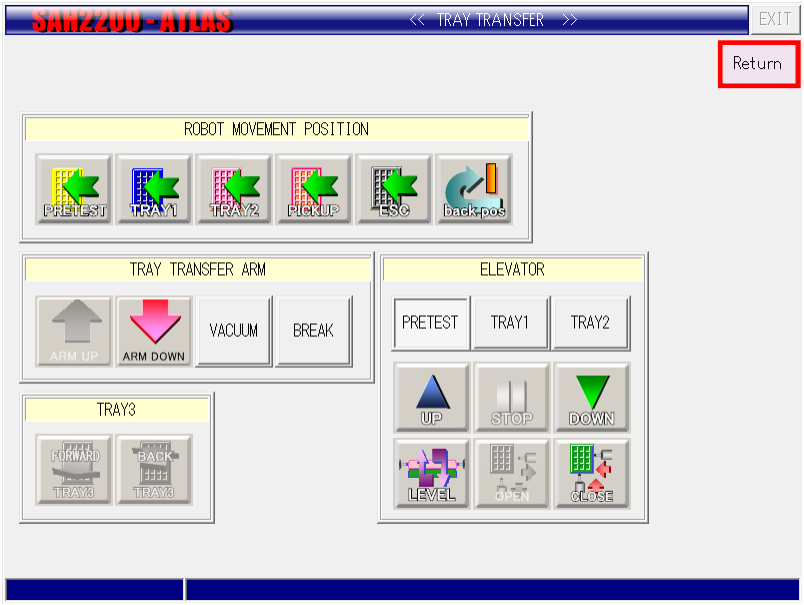


Fig 3.6F Manual tray transport screen

■PRETEST button: It will move to the PRETEST tray position.

■Tray1 button: It will move to the tray 1 position.

■Tray 2 button: It will move to the tray 2 position.

■Pickup button: It will move to the empty tray position.

■ESC button: It will move to the escape position.

■BACK POS button: It will return the tray transport arm to its original position.

(2)tray transfer arm

Make up and down motion of the tray transport arm.

■Arm up button: Tray transport arm will move to the top.

■Arm down button: Tray transport arm will move to the bottom.

■Vacuum button: The vacuum suction of the tray transport arm it will not be turned ON or OFF.

■Break button: The vacuum break of the tray transport arm it will not be turned ON or OFF.

(3)Tray 3

It will forward or backward movement of the tray 3 unit.

■Tray3 forward button: The tray 3 we will move forward.

■Tray3 backward button: The tray 3 we will move backward.

(4)Elevator

It will operate the PRETEST TRAY elevator and TRAY 1 elevator and TRAY 2 elevator individually.

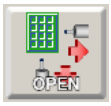
■Up button: It will move to above the elevator.

■Stop button: The elevator in operation it will be stopped.

■Down button: It will move the elevator down.

■Level button: It makes the leveling of the elevator.

■Chuck open button: It will open the elevator of the chuck.



■Chuck close button: It will close the elevator of the chuck.

(5) Manual menu screen is displayed when you press the **Return button**..
During automatic operation, the position of the tray transfer arm, the state of the transport arm, the position of the tray 3, and the vacuum state of the tray arm does not return to the original, not the **return button** is enabled.

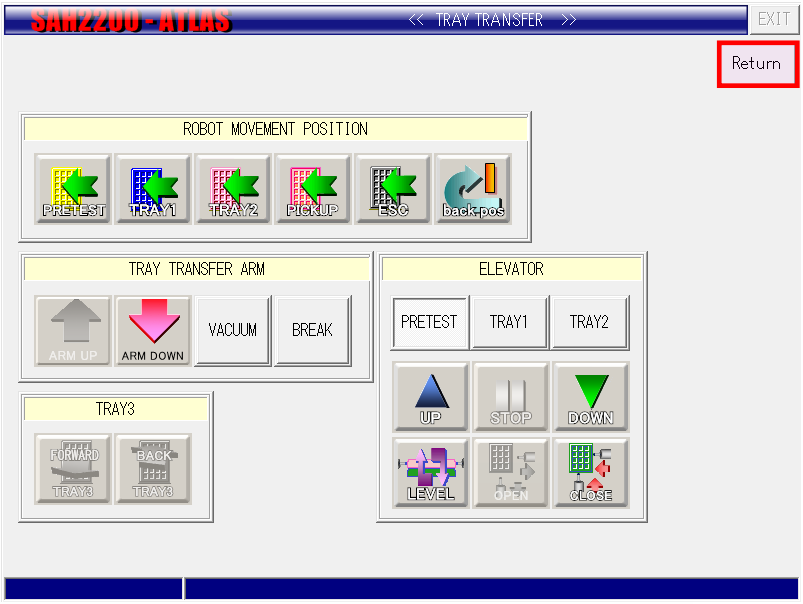


Fig 3.6F1 Manual tray transport screen

3.6.5 Manual Alignment

Only when the alignment enabled allows manual alignment operation.

(1) Make package alignment operation and recognition unit around the motor operation.

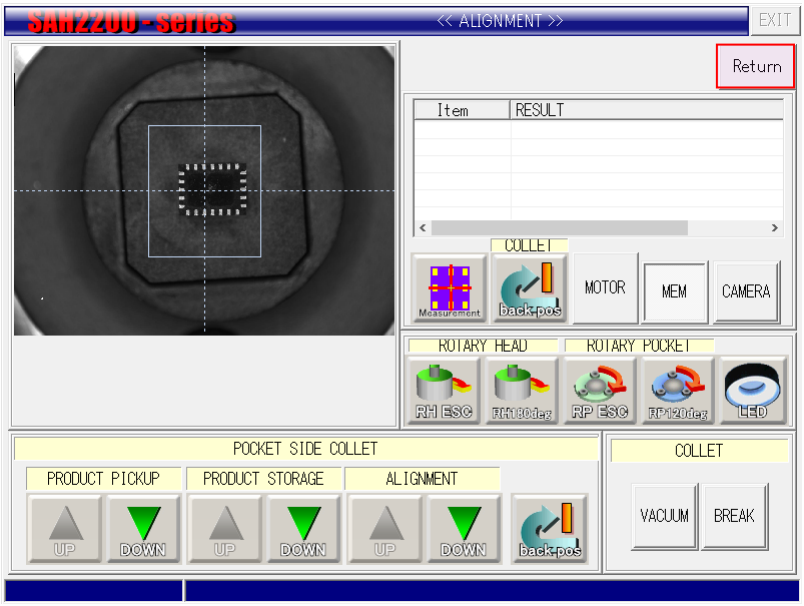
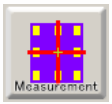


Fig 3.6G Manual alignment screen

■Measurement list: It will list the alignment measurement result.

■Measurement button: It makes the alignment operation.



■BACK POS Button: It will return the collet X-Y-θ motor to its original position.

■Motor operating button: During the alignment, it will do motor correction operation.

■Image button: It will display the captured image.

■Camera button: It will display the camera image.

■RH ESC button: it will move the rotary head to the escape position.

■RH 180 button: A rotary head rotate 180 degrees.

■RP ESC button: it will move the rotary pocket to the escape position.

■RP 120 button: A rotary pocket rotate 120 degrees.

2) It makes the vertical movement of the pocket side collet.

■PICKUP Up button: It will move the pocket side collet to the upper position.

■ PICKUP Down button: The pocket side collet move to pick-up position.

■REREASE Up button: It will move the pocket side collet to the upper position.

■RELEASE Down button:: The pocket side collet move to release position.

■PICTURE Up button: It will move the pocket side collet to the upper position.

■PICTURE Down button: The pocket side collet move to alignment position.

■BACK POS Button: It will move the pocket side collet to its original position.

(3) It makes the operation of the vacuum suction and vacuum breaking of the pocket side collet.

■Vacuum button: The vacuum suction of the pocket side collet will not be turned ON or OFF.

■Break button: The vacuum break of pocket side collet. it will not be turned ON or OFF.

(4)Manual menu screen is displayed when you press the **Return button**..

During automatic operation, the position of the pocket side collet, the position of the rotary head, the position of the rotary pocket, the position of the collet X-Y-Theta, if you do not return on the basis of the vacuum state of the pocket side collet, **return button** does not take effect.

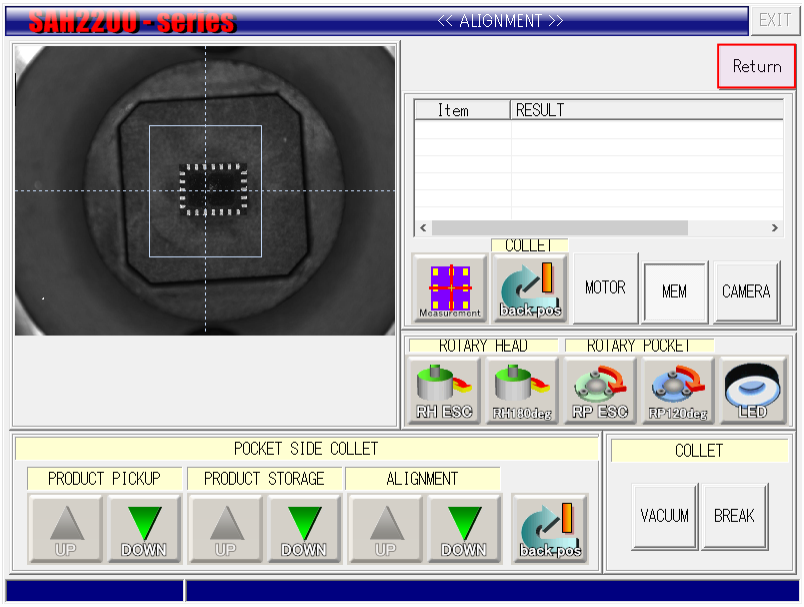


Fig 3.6G1 Manual alignment screen

3.6.6 Manual XY robot and hand robot

It will make the operation of the XY robot and hand robot.

(1)Hand A

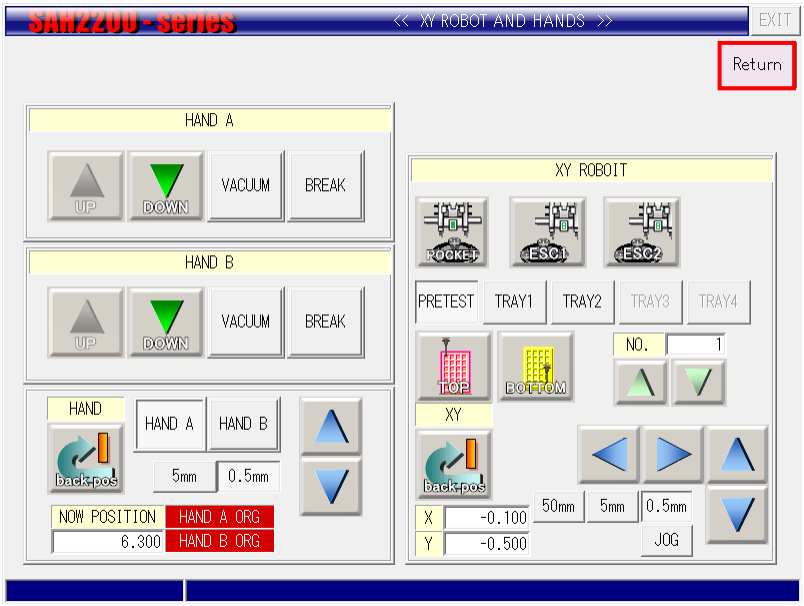


Fig 3.6H Manual XY robot and hand robot screen

■Up button: It will move the Hand A to the upper position.

■Down button: Hand A move to the lowered position of the pocket or tray.

■Vacuum button: The vacuum suction of the Hand A it will not be turned ON or OFF.

■Break button: The vacuum break of the Hand A it will not be turned ON or OFF.

(2)Hand B

■Up button: It will move the Hand B to the upper position.

■Down button: Hand B move to the lowered position of the pocket or tray.

■Vacuum button: The vacuum suction of the Hand B it will not be turned ON or OFF.

■Break button: The vacuum break of the Hand B it will not be turned ON or OFF.

(3)Hand operation panel

■Hand A: Choose the hand A.

■Hand B: Choose the hand B.

■Up button: It will scan move the specified hand in an upward direction.

■Down button: It will scan move the specified hand in a downward direction.

■ BACK POS Button: It will return the specified hand to its original position.

■Hand A origin sensor

It will show the state of the origin sensor of hand A. Red is the state of the origin sensor ON. Blue is the state of the origin sensor OFF.

■ Hand A origin sensor

It will show the state of the origin sensor of hand B. Red is the state of the origin sensor ON. Blue is the state of the origin sensor OFF.

(4)XY operation panel

It will operate the XY robot. At the origin sensor is in the ON state of the Hand A and Hand B, it is the XY robot can move. In a state in which the hand A or hand B is lowered, XY robot can not move.

■POCKET button: It will move to the pocket position.

■ESC1 button: It will move to the escape 1 position.

■ESC2 button: It will move to the escape 2 position.

■P TRAY button: Choose PRETEST tray.

■TRAY1 button: Choose Tray 1.

■ TRAY2 button : Choose Tray 2.

■ TRAY3 button : Choose Tray 3

■ TRAY4 button: Choose Tray 4

■TOP button: It will move to the top position of the selected tray.

■BOTTOM button: It will move to the final position of the selected tray.

■NO: It shows the movement position number of the tray.

■Count up button：The movement position of the tray is counted up, and then move to the next movement position.

■Count down button: Counts down the movement position of the tray, and then move to the front of the moving position.

■Left move button: Scan move the robot X to the left direction.

■Right move button: Scan move the robot X to the right direction.

■Front move button: Scan move the robot Y in the front direction.

■Back move button: Scan move the robot Y in the back direction.

■ BACK POS Button: It will return the robot XY to its original position.

(5)Manual menu screen is displayed when you press the Return button..

During automatic operation, the position of the robot XY, it is not returned to its original position and the vacuum state of the hand of the hand, a return button does not take effect.

3.6.7 Manual Pocket Collet

It will make the operation of the rotary pocket and collet.

(1)Socket side collet

It makes the operation of the socket side collet.

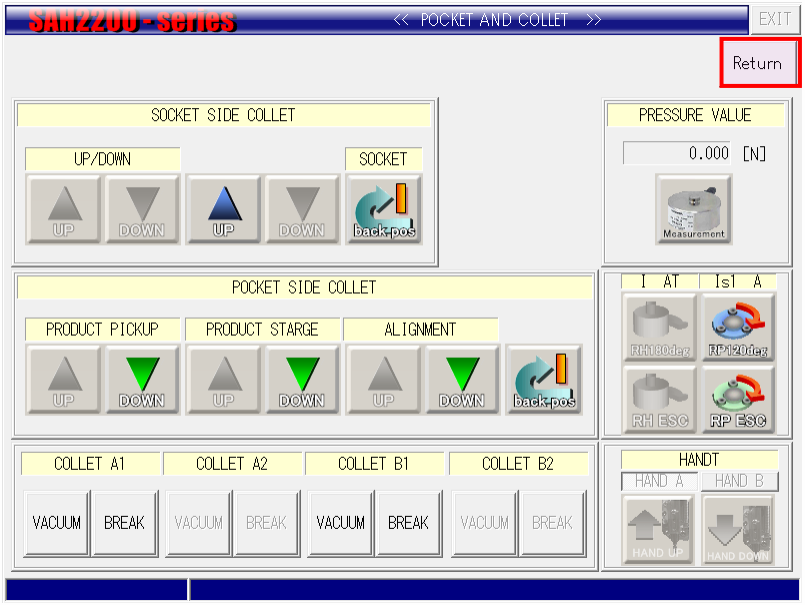


Fig 3.6I Manual Pocket Collet screen

■Up button: It will move the socket side collet to the upper position.

■Down button: It will move the socket side collet to drop position.

■Press up button: It will raise move the socket side collet to the drop position.

■Press down button: It will move the socket side collet to press position.

■BACK POS Button: It will return the socket side collet to its original position.

(2)Pocket side collet

It makes the operation of the pocket side collet.

■PICKUP Up button: It will move the pocket side collet to the upper position.

■PICKUP Down button: The pocket side collet move to pick-up position.

■RELEASE Up button: It will move the pocket side collet to the upper position.

■RELEASE Down button: The pocket side collet move to release position.

■PICTURE Up button:: It will move the pocket side collet to the upper position.

■PICTUTE Down button: The pocket side collet move to alignment position.

■BACK POS Button: It will move the pocket side collet to its original position.

(3)Collet adsorption

It makes the operation of the vacuum suction and vacuum breaking of the collet.

■A1 Vacuum button: The vacuum suction of the pocket side collet A1 will not be turned ON or OFF.

■A1 Break button: The vacuum break of pocket side collet A1. it will not be turned ON or OFF.

■A2 Vacuum button: The vacuum suction of the pocket side collet A2 will not be turned ON or OFF.

■A2 Break button: The vacuum break of pocket side collet A2. it will not be turned ON or OFF.

■B1 Vacuum button: The vacuum suction of the pocket side collet B1 will not be turned ON or OFF.

■B1 Break button: The vacuum break of pocket side collet B1. it will not be turned ON or OFF.

■B2 Vacuum button: The vacuum suction of the pocket side collet B2 will not be turned ON or OFF.

■B2 Break button: The vacuum break of pocket side collet B2. it will not be turned ON or OFF.

(4)Pressure measurement

■PRESSURE CLC button: If the pressure monitor measurement is enabled, if the socket side collet is in a pressure position, it will measure the applied pressure.

(5)Rotary head and rotary pocket operation

■RH ESC button: it will move the rotary head to the escape position.

■RH 180 button: A rotary head rotate 180 degrees.

■RP ESC button: it will move the rotary pocket to the escape position.

■RP 120 button: A rotary pocket rotate 120 degrees.

(6)Hand

When operating the pocket and collet manually pocket collet screen, on a rotary pocket, it is valid operation when the hand is in the lowered position. If Hand A and Hand B is raised position, the operation of the hand is invalid.

■Hand up button: : It will move the hand to the upper position.

■Hand down button: It will return the hand to the down position.

(7)Manual menu screen is displayed when you press the **Return button**..

During automatic operation, the position of the socket side collet, the position of the pocket side collet, the position of the rotary pocket, the position of the rotary head, and not returned to the original position and the vacuum state of the collet of the hand, a **return button** does not take effect.

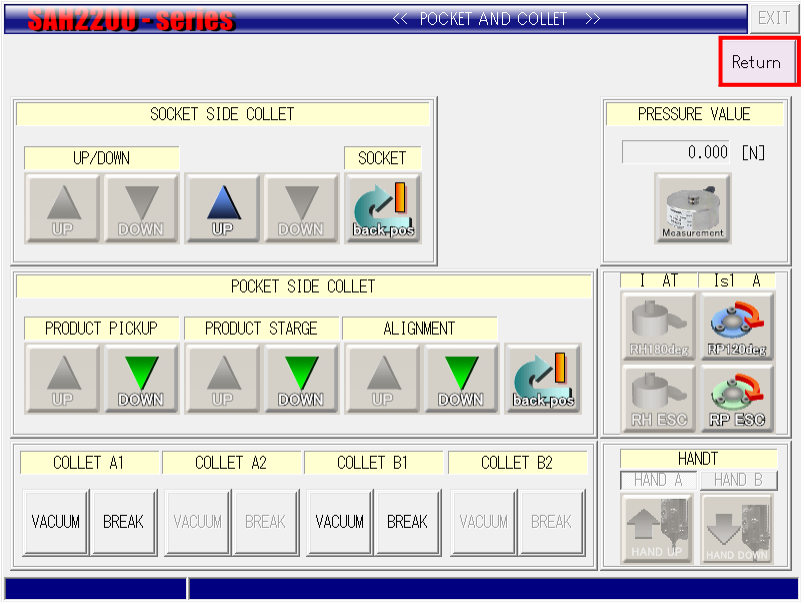


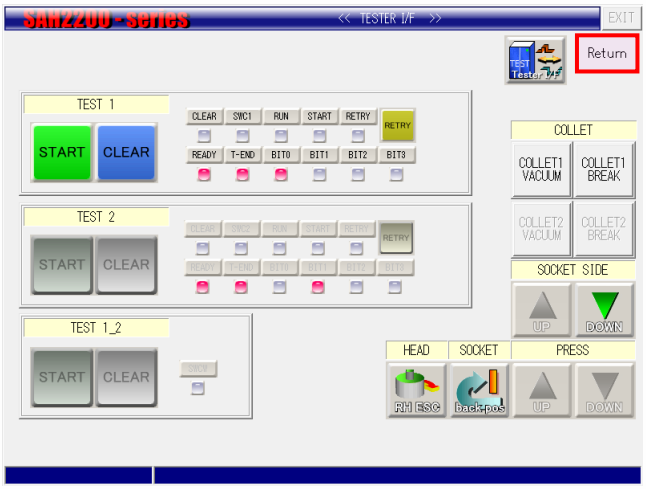
Fig 3.6I1 Manual Pocket Collet screen

3.6.8 Manual Test

It makes the operation of the manual test measurement.

(1)Test 1

It makes the measurement operation of the socket 1 side.



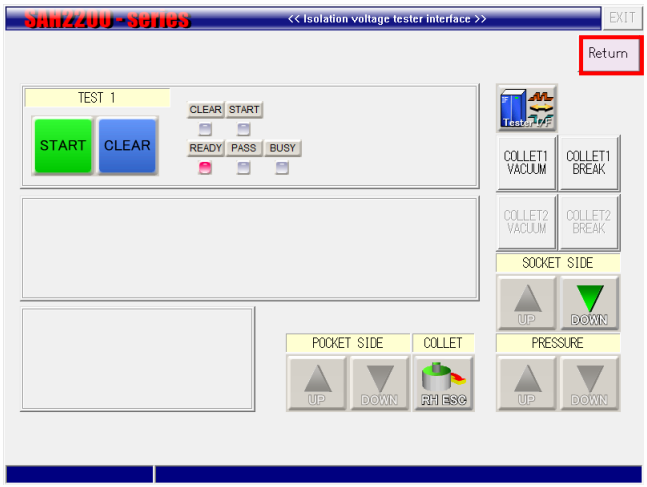


Fig 3.6J Manual test screen

■Start button: It will start the test measurement.

■Clear button: It clears the end signal and category signal.

■Retry button: If retry signal is active, it will turn ON or OFF the retry signal.

(2)Test 2

It makes the measurement operation of the socket 2 side.

■Start button: It will start the test measurement.

■Clear button: It clears the end signal and category signal.

■Retry button: If retry signal is active, it will turn ON or OFF the retry signal.

(3)Test 1,2

It makes the measurement operation of the socket 1 and socket 2.

■Start button: It will start the test measurement.

■Clear button: It clears the end signal and category signal.

■Test I/F button: It will open the test I / F setting screen.

(4)Pocket side collet

It makes the operation of the pocket side collet.

■PICKUP Up button: It will move the pocket side collet to the upper position.

■PICKUP Down button: The pocket side collet move to pick-up position.

(5)Rotary head

■RH ESC button: it will move the rotary head to the escape position.

(6) Socket adsorption

It makes the operation of the vacuum suction and vacuum breaking of the socket side collet.

■collet 1 Vacuum button: The vacuum suction of the socket side collet 1 will not be turned ON or OFF.

■collet 1 Break button: The vacuum break of socket side collet 1. it will not be turned ON or OFF.

■collet 2 Vacuum button: The vacuum suction of the socket side collet 2 will not be turned ON or OFF.

■collet 2 Break button: The vacuum break of socket side collet 2. it will not be turned ON or OFF.

(7)Socket side collet

It makes the operation of the socket side collet.

■Up button: It will move the socket side collet to the upper position.

■Down button: It will move the socket side collet to drop position.

■Press up button: It will raise move the socket side collet to the drop position.

■Press down button: It will move the socket side collet to press position.

(8)Manual menu screen is displayed when you press the Return button..

During automatic operation, the position of the socket side collet, the position of the pocket side collet, the position of the rotary head, and do not return based on the vacuum state of the collet, return button does not take effect.

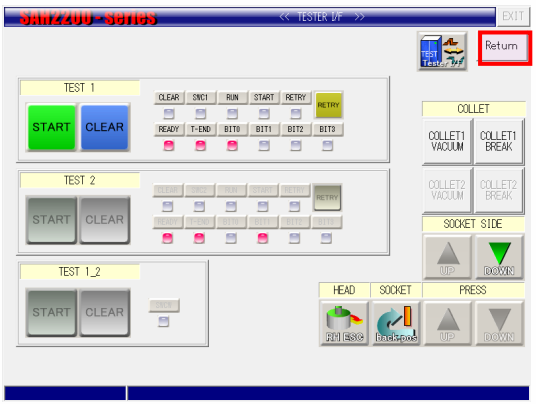


Fig 3.6J1 Manual test screen

3.6.9 Operating time measurement

It will measure the operating time of the hand and the collet and the chuck. Press the **Time movement button** from manual operation menu screen of FIG. 3.6B.

Fig 3.6K Operating time measurement screen

(1)Pocket side collet operating time measurement

It will measure the operating time of the pocket side collet.

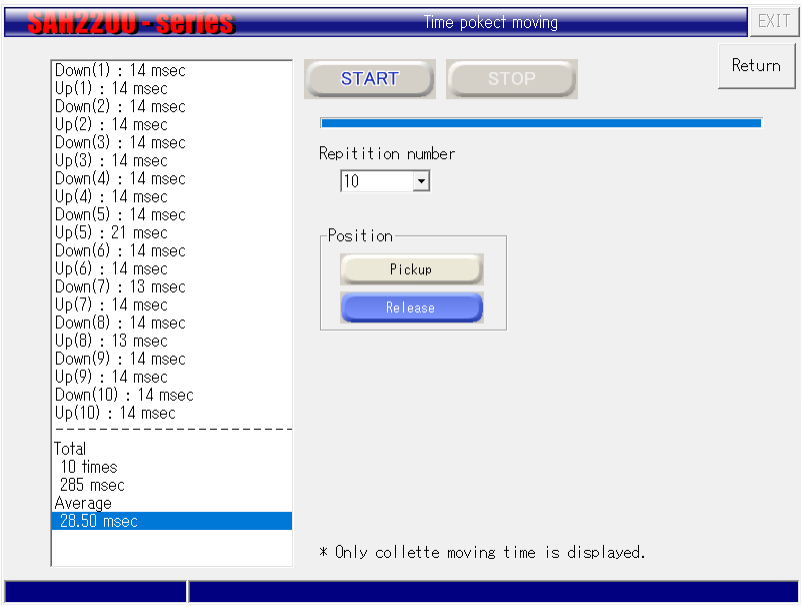


Fig 3.6L Pocket side collet operating time measurement screen

■Start button

It will start the operation measurement.

■Stop button

It will suspend the operation measurement.

■Number of repetition

You set the number of repeat test. Standard is 10 times.

■Pickup

It will select the pickup position.

■Release

It will select the storage position.

(2)Socket side collet operating time measurement

It will measure the operating time of the socket side collet.

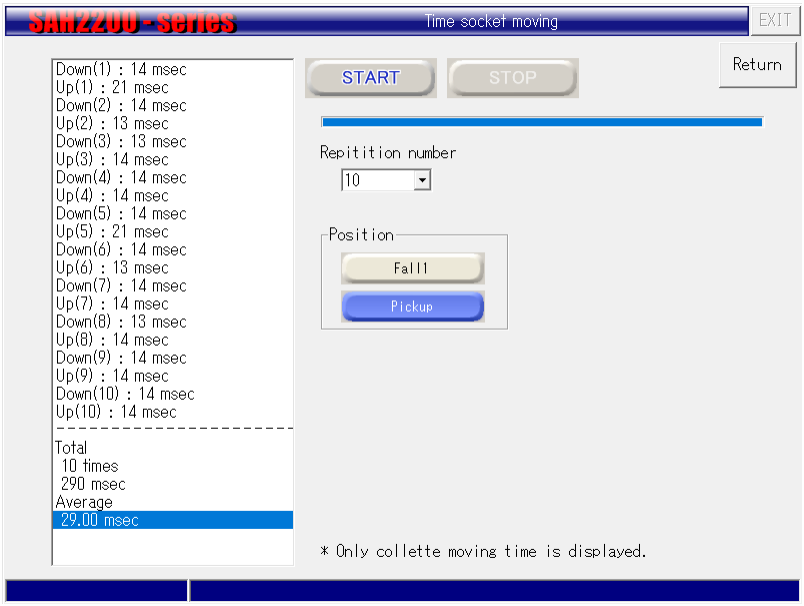


Fig 3.6M Socket side collet operating time measurement screen

■Start button

It will start the operation measurement.

■Stop button

It will suspend the operation measurement.

■Number of repetition

You set the number of repeat test. Standard is 10 times.

■Fail 1

It will select the fail 1 position.

■Pickup

It will select the press position.

(3)Hand A operating time measurement

It will measure the operating time of the Hand A.

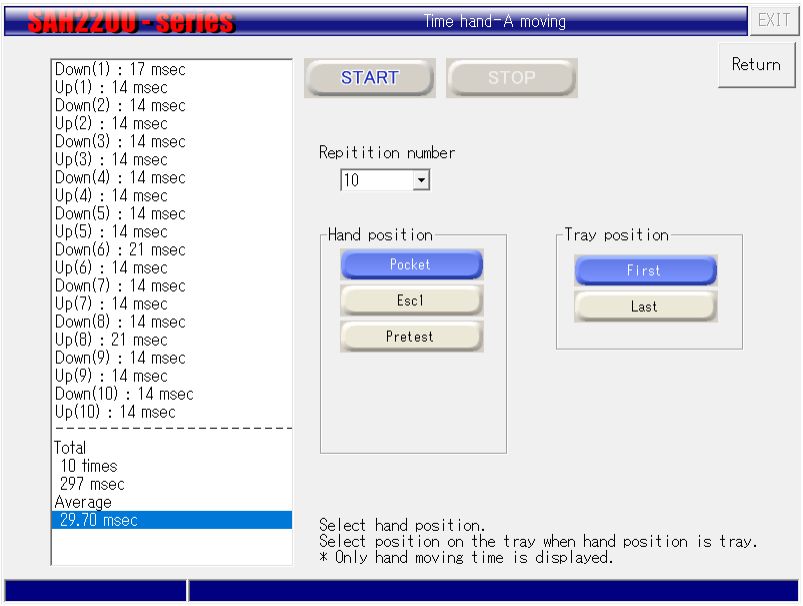


Fig 3.6N Hand A operating time measurement screen

■Start button

It will start the operation measurement.

■Stop button

It will suspend the operation measurement.

■Number of repetition

You set the number of repeat test. Standard is 10 times.

■Pocket

It will select the pocket position.

■ESC 1

It will select the escape 1 position.

■Pretest

It will select the pretest tray.

■First

It will select the first position of the pretest tray.

■Last

It will select the final position of the pretest tray.

(4)Hand B operating time measurement

It will measure the operating time of the Hand B.

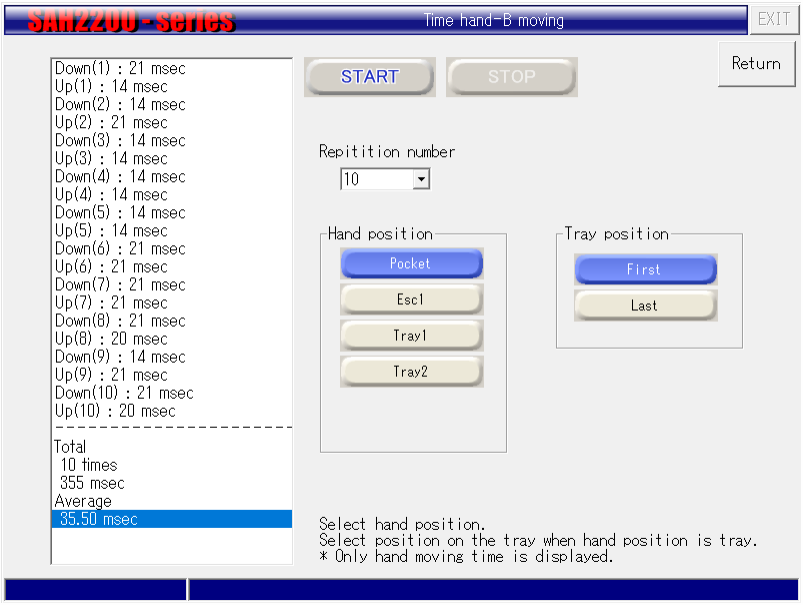


Fig 3.6O Hand B operating time measurement screen

■Start button

It will start the operation measurement.

■Stop button

It will suspend the operation measurement.

■Number of repetition

You set the number of repeat test. Standard is 10 times.

■Pocket

It will select the pocket position.

■ESC 1

It will select the escape 1 position.

■Tray 1

It will select the tray 1.

■Tray 2

It will select the tray 2..

■First

It will select the first position of the tray.

■Last

It will select the final position of the tray.

(5)Chuck operating time measurement

It will measure the operating time of the Chuck.

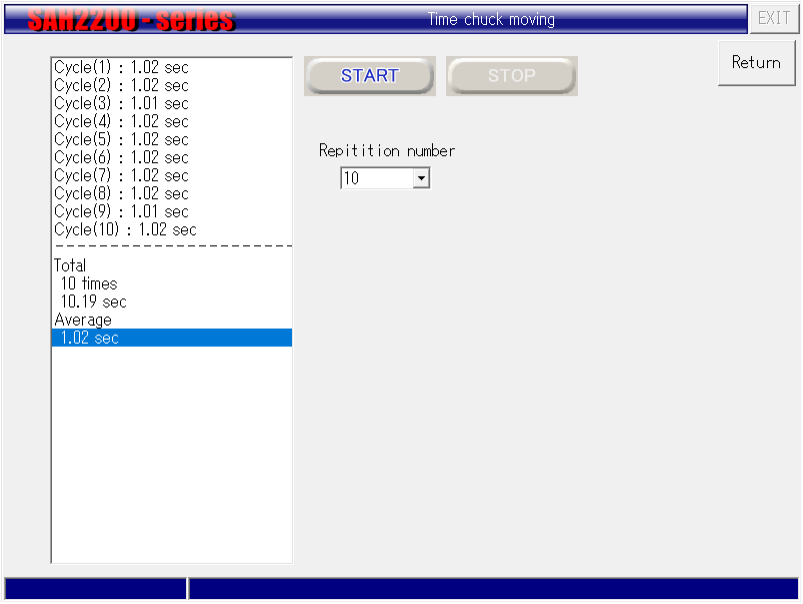


Fig 3.6P Chuck operating time measurement screen

■Start button

It will start the operation measurement.

■Stop button

It will suspend the operation measurement.

■Number of repetition

You set the number of repeat test. Standard is 10 times.

■Normal

It will repeat the chuck opening operation and the chuck closing operation.

■Cycle

It will perform the chuck 1 cycle operation.

(6)Tray arm operating time measurement

It will measure the operating time of the tray arm(up / down).

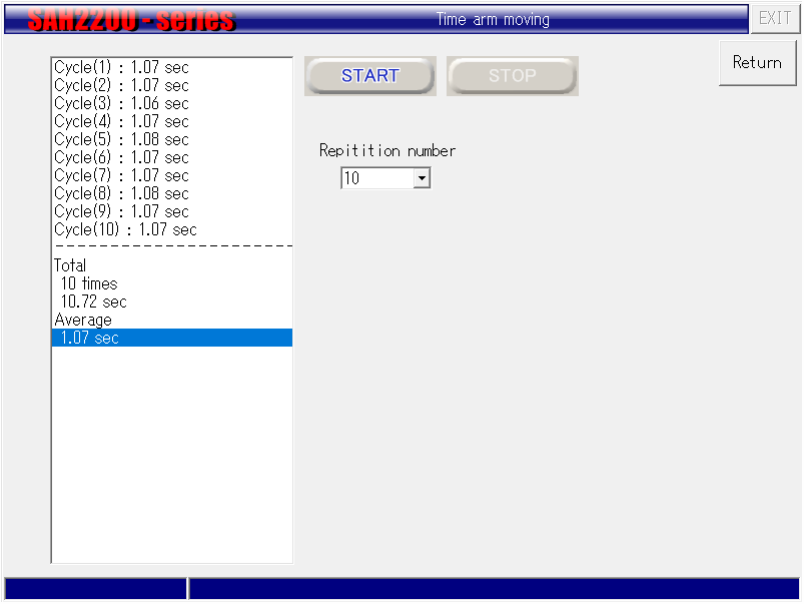


Fig 3.6Q Tray arm operating time measurement screen

■Start button

It will start the operation measurement.

■Stop button

It will suspend the operation measurement.

4.Error

4.1 Error

4.1. 1 Processing in the event of an error

If an error occurs for some reason it shifts to the error stop screen.

If an error stop screen is displayed, after the occurrence due correctly Processing of error, to close the error stop screen by pressing the **reset button**. Errors are classified into two types by the cause.

It shows a list of each classification in the next section.

Fig 4.1A Error stop screen

4.1.2 Error Recovery in reset button and start button

Error name	Cause	Input IO	Normal status	Release method
Cover open	Cover is open.	Cover open	ON	Close the cover
Elevator door open	Elevator door is open.	Elevator door open	ON	Close the elevator door
Primary air pressure too low error	Primary air supply pressure has not reached the specified value.	Primary air pressure OK	ON	confirm that the pressure of supply dry air is within the set value
Tray transfer arm up timeout error	Tray transport raising operation has timed out.	Tray transfer upper limit	ON	a) Adjustment of the speed controller b) Check of the time-out timer
Tray transfer arm down timeout error	Tray transport downward movement has timed out.	Tray transfer upper limit Tray transfer landing check	OFF ON	a) Adjustment of the speed controller b) Check of the time-out timer
Tray transfer arm down error	Upper limit sensor of the tray arm is OFF.	Tray transfer upper limit	ON	Checking the upper limit sensor
Pretest tray suction error	It failed to adsorption the pretest tray.	Tray suction OK	ON	Adjust the vacuum air, the tray is adsorbed.
Pretest Tray drop error	The pretest tray during transport, the transport arm was dropped tray.	Tray suction OK	ON	It adsorbs the dropped tray on the arm
Tray suction error	Was empty tray buffer tray taken out, but it could not be taken out by the transfer arm.	Tray suction OK	ON	It adsorbs the tray
Tray drop error	During the transport of the empty tray, it was dropped the tray	Tray suction OK	ON	It adsorbs the dropped tray on the arm
Robot hand A product pickup error	Hand A is taken out from the tray of the products, hand A did not adsorb the product.	Robot hand A suction OK	ON	a)Check whether there is product in the tray b) check the product pick-up position c)Adjust the vacuum air
Robot hand A product drop error	Hand A s in transport, hand A was drop the product.	Robot hand A suction OK	ON	It adsorbs the fallen product in hand

Error Recovery in reset button and start button (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Robot hand A product pickup error(Hot plate)	Hand A is taken out from the plate of the products, hand A did not adsorb the product.	Robot hand A suction OK	ON	a)Check whether there is product in the plate b) check the product pick-up position c)Adjust the vacuum air
Robot hand A product pickup error(Pocket)	Hand A is taken out from the loader rotary pocket of the products, hand A did not adsorb the product.	Robot hand A suction OK	ON	a) Check the status of the product b)Adjust the vacuum air
Robot hand B product pickup error	Hand B is taken out of the product from the pocket, but the hand B did not adsorb the product.	Robot hand B suction OK	ON	a)Check the status of the product. c)Adjust the vacuum air
Robot hand B drop error	Robot hand B is in the transport, it was allowed to drop the product	Robot hand B suction OK	ON	It adsorbs the product is dropped on the hand B.
Robot hand B drop error (rotary pocket)	Robot hand B is during transportation from the rotating pocket, was dropped the product.	Robot hand B suction OK	ON	It adsorbs the product is dropped on the hand B.
Robot hand B product pickup error (rotary pocket)	Although hand-B from the rotation pocket was taken out of the product, hand B did not adsorb the product.	Robot hand B suction OK	ON	a)Check the status of the product. c)Adjust the vacuum air
Robot hand B product loading error	Hand B did not properly storing the product to pocket A.	Robot hand B suction OK	ON	It stores the product in the pocket A.
Pocket A IC remain error	Residual product is present in the pocket B at the time of automatic operation start.	Product present in pocket A.	ON	Remove the product from the pocket A
Pocket B IC remain error	Residual product is present in the pocket B at the time of automatic operation start.	Product present in pocket B. Pocket B product setting error	ON ON	Remove the product from the pocket B
Pocket C IC remain error	Residual product is present in the pocket C at the time of automatic operation start.	Product present in pocket C.	ON	Remove the product from the pocket C
Socket exist sensor error	Residual product is present in the socket when automatic operation start.	Product present in socket. Socket product setting error 1. Socket product setting error A2.	ON ON ON	Remove the product from the socket

Error Recovery in reset button and start button (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Socket exist sensor error	Residual product is present in the socket when automatic operation start.	Product present in socket. Socket product setting error 1. Socket product setting error A2.	ON ON ON	Remove the product from the socket
Please checks the Residual goods.	Please check whether socket or socket in there is a residual product. (Warning message)			It will check the status of the socket.
Ion blowing error	Ion blow error has occurred.	Ion blowing abnormal	ON OFF	a)check the power status of the Ion blow. b) It will check the status of the Ion blow inside of contact
Pretest IC count over error	It is reached to the specified construction schedule number, but there are still supply products.			Set the number of construction plans again
Product present in pocket A error	Product exists that would not in the pocket A	Product present in pocket A.	ON	Remove a product from the pocket A
Robot hand A product loading error	Hand A is accommodating the product to pocket A, but the hand A could not be stored in the product	Product present in pocket A.	OFF	It stores the product in the pocket A.
Pocket B product loading error	Product is not stored properly in the pocket B.	Pocket B product setting error	ON	Correctly set the product to pocket B
Pocket B product absent error	No product is should be in the pocket B.	Product present in pocket B.	OFF	Correctly set the product to pocket B
Pocket side collet A product suction error	It was product of adsorption in the pocket side collet A but, Colette did not adsorb the product.	Collet A1 suction OK	ON	Correctly set the product to pocket B
Pocket side collet B product suction error	It was product of adsorption in the pocket side collet B but, Colette did not adsorb the product.	Collet B1 suction OK	ON	Correctly set the product to pocket B
Pocket side collet A product pickup error	It was pick up the product on a pocket side collet A, but could not adsorb the product.	Collet A1 suction OK Pocket B product setting error Product present in pocket B.	ON ON OFF	It will check the status of the product of pocket B.
Pocket side collet B product pickup error	It was pick up the product on a pocket side collet B, but could not adsorb the product.	Collet B1 suction OK Pocket B product setting error Product present in pocket B.	ON ON OFF	It will check the status of the product of pocket B.

Error Recovery in reset button and start button (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Pocket side collet A product loading error	Pocket side collet A was not able to store the product in rotation pocket.	Pocket B product setting error	ON	It will check the status of the product of the rotating pocket
Pocket side collet B product loading error	Pocket side collet B was not able to store the product in rotation pocket.	Pocket B product setting error	ON	It will check the status of the product of the rotating pocket
Product present in pocket C error	Product exists that would not in the pocket C	Product present in pocket C	ON	It removes the product from the pocket C.
Pocket side collet drop error	Rotary head during rotation, pocket side collet was dropped product	Product present in pocket C	ON	It adsorbs the fallen product to pocket side collet
Socket side collet drop error	Rotary head during rotation, socket side collet was dropped product	Collet A1 (B1) suction OK	ON	It adsorbs the fallen product to socket side collet
Socket product loading error	Product has not been stored properly in the socket.	Socket product setting error 1. Socket product setting error A2.	ON ON	Correctly set the product to the socket
Socket product absent error	No product is stored in the socket.	Product present in socket.	OFF	Correctly set the product to the socket
Socket side collet product suction error	It could not adsorb the product on a socket side collet.	Collet A1 (B1) suction OK	ON	The product of the socket it is correctly adsorption.
Please remove products of collet A	There is a residual product in the socket side collet A.	Product present in pocket B. Pocket B product setting error	ON ON	Remove the product of the socket
Please remove products of collet B	There is a residual product in the socket side collet B.	Product present in pocket B. Pocket B product setting error	ON ON	Remove the product of the socket
Communication error	Test results, it received the BIN signal that is not specified.			a)check the contact of the tester I / F cable b) Check the tester I / F setting
Continuous defect detection error	Reaching the number of defective products are specified consecutively.			Please clean the socket.

Error Recovery in reset button and start button (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Tester error.	Tester error has occurred in the laser mark communication.			
Yield abnormal.	Yield abnormal error has occurred.			
Test1 clear error.	It failed to clear the end-1 signal.	END1	OFF	Connector connection check of the internal I / F board
Test2 clear error.	It failed to clear the end-2 signal.	END2	OFF	Connector connection check of the internal I / F board
Test1 timeout error	End 1 signal did not turn ON within a predetermined period of time.	END1	ON	Connection check of the measurement cable
Test2 timeout error	End 2 signal did not turn ON within a predetermined period of time.	END2	ON	Connection check of the measurement cable
Test timeout error	End signal did not turn ON within a predetermined period of time.	END1/END2	ON	Connection check of the measurement cable
Error : Tester Alarm	Tester alarm error has occurred in the GPIB communication.			Connection check of GPIB communication cable
GPIB communication error	In GPIB communication, communication error has occurred.			Connection check of GPIB communication cable
90 rotation cylinder timeout error	Operation of the rotary cylinder has timed out.			
Loader Rotary pocket exist sensor error	There is a residual product to the loader rotation pocket.			Remove the product of the rotation pocket.
Loader Rotary pocket absence error	Product could not be stored in the loader rotation pocket.			It will properly store the product to the rotating pocket.
Unloader Rotary pocket exist sensor error	There is a residual product to the unloader rotation pocket.			Remove the product of the rotation pocket.
unloader Rotary pocket absence error	Product could not be stored in the unloader rotation pocket.			It will properly store the product to the rotating pocket.
Press monitor communication timeout error.	Pressure measurement is not completed in the time-out.			a)Check Connection of communication cable b) Check the setting value of the amplifier unit

Error Recovery in reset button and start button (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Press monitor communication error(1)	Un executable condition has occurred.			a)Check Connection of communication cable b) Check the setting value of the amplifier unit
Press monitor communication error(2)	Errors due to the convenience of the amplifier unit has occurred.			a)Check Connection of communication cable b) Check the setting value of the amplifier unit
Press monitor parity error 10.	Parity error has occurred.			a)Check Connection of communication cable b) Check the setting value of the amplifier unit
Press monitor framing error 11.	Framing error has occurred.			a)Check Connection of communication cable b) Check the setting value of the amplifier unit
Press monitor over run error 12.	Overrun error has occurred.			a)Check Connection of communication cable b) Check the setting value of the amplifier unit
Press monitor date length error 13.	Data code or data length error has occurred.			a)Check Connection of communication cable b) Check the setting value of the amplifier unit
Press monitor no command error 14.	No command error has occurred.			a)Check Connection of communication cable b) Check the setting value of the amplifier unit
Press monitor max limit.	Measured value of the pressure monitor has exceeded the upper limit			a)Check if the product is stored on a socket b) Review of the limit value
Press monitor min limit.	Measured value of the pressure monitor is less than the lower limit.			a)Check if the product is stored on a socket b) Review of the limit value
Start script error	Start script execution error in the event of a network script enabled			Check the script batch file
End script error	End script execution error in the event of a network script enabled			Check the script batch file

Error Recovery in reset button and start button (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Pretest tray level sensor error	XY robot can not move for pretest tray level sensor is OFF.	Pretest tray elevator level	OFF	It will check whether there is no obstacle on the level sensor of the elevator
Tray1 level sensor error	XY robot can not move for tray 1 level sensor is OFF.	Tray1 elevator level	OFF	It will check whether there is no obstacle on the level sensor of the elevator
Tray2 level sensor error	XY robot can not move for tray 2 level sensor is OFF.	Tray2 elevator level	OFF	It will check whether there is no obstacle on the level sensor of the elevator
Hand A vacuum sensor error	During the initial adsorption check, vacuum sensor of the hand is ON.	Robot hand A suction OK	OFF	a)Check the state of the nozzle b) Status check of the vacuum pump
Hand B vacuum sensor error	During the initial adsorption check, vacuum sensor of the hand is ON.	Robot hand B suction OK	OFF	a)Check the state of the nozzle b) Status check of the vacuum pump
A1 collet vacuum sensor error	During the initial adsorption check, vacuum sensor of the collet is ON.	Collet A1 suction OK	OFF	a)Check the state of the nozzle b) Status check of the vacuum pump
A2 collet vacuum sensor error	During the initial adsorption check, vacuum sensor of the collet is ON.	Collet A2 suction OK	OFF	a)Check the state of the nozzle b) Status check of the vacuum pump
B1 collet vacuum sensor error	During the initial adsorption check, vacuum sensor of the collet is ON.	Collet B1 suction OK	OFF	a)Check the state of the nozzle b) Status check of the vacuum pump
B2 collet vacuum sensor error	During the initial adsorption check, vacuum sensor of the collet is ON.	Collet B1 suction OK	OFF	a)Check the state of the nozzle b) Status check of the vacuum pump
Socket management system stop	It was stopped by the socket management system.			Check the status of the socket
Socket management system warning	Warning from the socket management system was there.			Check the status of the socket
The temperature doesn't reach a set value.	Hot plate has not been heated to a set temperature.			Wait until the temperature rises

Error Recovery in reset button and start button (Continuation)

Error name	Cause	Input IO	Normal status	Release method
The Collet A temperature doesn't reach a set value.	The collet A has not been heated to a set temperature.			Wait until the temperature rises
The Collet B temperature doesn't reach a set value.	The collet B has not been heated to a set temperature.			Wait until the temperature rises
The Socket 1 temperature doesn't reach a set value.	The socket 1 has not been heated to a set temperature.			Wait until the temperature rises
The Socket 2 temperature doesn't reach a set value.	The socket 2 has not been heated to a set temperature.			Wait until the temperature rises
Please wait.	This is the state of the temperature stabilization wait after setting the temperature rise.			Wait until the temperature is stable
Alignment error.	Alignment error occurs set number of times continuously at the time of alignment enabled.			a)Check the alignment image b) Check the alignment teaching conditions
USB optical unit position error.	USB camera unit is not in a normal position.	USB Camera unit back side limit	ON	a)It returned to the original position of the USB camera unit. b) Check of the sensor
Door lock close error	When the door lock closed, the sensor is not ON.	Door lock check	ON	a) Checking the timer b) Checking the timer
Door lock open error	When the door lock opening, the sensor is not turned OFF.	Door lock check	OFF	a) Checking the timer b) Checking the timer
LAN Connect error	During the test log file effective, it failed to log destination connected by a LAN via during online test.			a)Check the LAN cable b) Check the destination folder
Tray IC miss position and please remove the IC!	Product was not stored properly in the tray.	TRAY STORAGE SENSOR	OFF	Remove the product from the tray

4.1.3 Error to return at the origin return operation

Error name	Cause	Input IO	Normal status	Release method
Emergency stop	Emergency stop switch is pressed	Emergency stop	ON	Please remove the emergency stop factors to release the emergency stop switch
Pretest elevator origin adjustment timeout error	Origin return of the elevator has timed out.	Pretest tray elevator lower limit	ON	a)Remove the conditions that I interfere with the downward movement of the elevator b) Please do check the timer c) The adjustment of the sensor
Tray1 elevator origin adjustment timeout error	Origin return of the elevator has timed out.	Tray1 elevator lower limit	ON	a)Remove the conditions that I interfere with the downward movement of the elevator b) Please do check the timer c) The adjustment of the sensor
Tray2 elevator origin adjustment timeout error	Origin return of the elevator has timed out.	Tray2 elevator lower limit	ON	a)Remove the conditions that I interfere with the downward movement of the elevator b) Please do check the timer c) The adjustment of the sensor
Robot hand A origin adjustment timeout error	Returns to origin of hand A has timed out.			a) Check the top position b) Checking the motor parameters c) Checking the timer d) Check the origin sensor
Robot hand B origin adjustment timeout error	Returns to origin of hand B has timed out.			a) Check the top position b) Checking the motor parameters c) Checking the timer d) Check the origin sensor
XY robot origin adjustment timeout error	Origin return of the XY robot has timed out.			a) Check the top position b) Checking the motor parameters c) Checking the timer d) Check the origin sensor
Pocket side collet position error	It can not be the operation of the rotary head and rotary pocket for pocket side collet is lowering	a)Pocket side Collet upper check b)Origin sensor	ON ON	a) Status check of the sensor b) If the collet is lowering, after an emergency stop, go origin return
Socket side collet position error	It can not be the operation of the rotary head and rotary pocket for socket side collet is lowering	a)Socket side Collet upper check b)Origin sensor	ON ON	a) Status check of the sensor b) If the collet is lowering, after an emergency stop, go origin return

Error to return at the origin return operation (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Pocket origin adjustment timeout error	Origin return operation of the rotary pocket has timed out			a)Checking the motor parameters b) Checking the timer c) Check the origin sensor
Collet A origin adjustment timeout error	Origin return operation of the collet A has timed out			a)Checking the motor parameters b) Checking the timer c) Check the origin sensor
Collet B origin adjustment timeout error	Origin return operation of the collet B has timed out			a)Checking the motor parameters b) Checking the timer c) Check the origin sensor
Rotary head origin adjustment timeout error	Origin return operation of the rotary head has timed out			a)Checking the motor parameters b) Checking the timer c) Check the origin sensor
Tray transfer arm origin adjustment timeout error	Origin return operation of the tray transfer arm has timed out			a)Checking the motor parameters b) Checking the timer c) Check the origin sensor
Tray transfer arm position error	During the Returns to origin of the tray transfer arm, the position of the tray transport arm is invalid.			Please correct the position of the arm.
Tray arm position error	When the robot XY movement, the position of the tray transport arm is invalid.			After an emergency stop, Please correct the position of the arm. Please go origin return again
Robot X limit error	When the software origin move, it was stopped because the limit sensor is ON			Check the soft limit position
Robot Y limit error	When the software origin move, it was stopped because the limit sensor is ON			Check the soft limit position
Hand A limit error	When the software origin move, it was stopped because the limit sensor is ON			Check the soft limit position
Hand B limit error	When the software origin move, it was stopped because the limit sensor is ON			Check the soft limit position
Pocket pickup limit error	When the software origin move, it was stopped because the limit sensor is ON			Check the soft limit position

Error to return at the origin return operation (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Socket pickup limit error	When the software origin move, it was stopped because the limit sensor is ON			Check the soft limit position
Tray transfer limit error	When the software origin move, it was stopped because the limit sensor is ON			Check the soft limit position
Pretest tray operation timeout error	Pretest tray elevator operation has timed out.			a)Please check if there is any obstacle on the elevator b) Checking the timer
Tray 1 operation timeout error	Tray 1 elevator operation has timed out.			a)Please check if there is any obstacle on the elevator b) Checking the timer
Tray 2 operation timeout error	Tray 2 elevator operation has timed out.			a)Please check if there is any obstacle on the elevator b) Checking the timer
Pretest tray leveling timeout error	Level operation of pretest tray has timed out			Check the set state of the tray
Tray 1 leveling timeout error	Level operation of tray 1 has timed out			Check the set state of the tray
Tray 2 leveling timeout error	Level operation of tray 2 has timed out			Check the set state of the tray
Tray transfer arm motor operation timeout error	Operation of moving tray transport arm has timed out.			a)Check the position data b)Checking the motor parameters c) Checking the timer
XY Robot motor operation timeout error	Operation of moving robot XY times out。			a)Check the position data b)Checking the motor parameters c) Checking the timer
XY Robot condition abnormal error	For the reasons Hand A or hand B is lowering, it is not possible to move the robot XY.			a)Check the origin sensor of hand b) After an emergency stop, it does Origin return.

Error to return at the origin return operation (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Robot hand A up timeout error	Upward movement of the hand A has timed out			a)Check the obstacles that are preventing the upward movement b) Checking the motor parameters c) Checking the timer d) After checking the origin sensor, it does an emergency stop. Please then go home return operation
Robot hand A down timeout error	Downward movement of the hand A has timed out			a) Checking the motor parameters b) Checking the timer c) After checking the origin sensor, it does an emergency stop. Please then go home return operation
Robot hand A timeout error	Operation of moving the hand A times out. (Manual operation)			a) Checking the motor parameters b) Checking the timer c) After checking the origin sensor, it does an emergency stop. Please then go home return operation
Robot hand B up timeout error	Upward movement of the hand B has timed out			a)Check the obstacles that are preventing the upward movement b) Checking the motor parameters c) Checking the timer d) After checking the origin sensor, it does an emergency stop. Please then go home return operation
Robot hand B product pickup error	Downward movement of the hand B has timed out			a) Checking the motor parameters b) Checking the timer c) After checking the origin sensor, it does an emergency stop. Please then go home return operation
Robot hand B timeout error	Operation of moving the hand B times out. (Manual operation)			a) Checking the motor parameters b) Checking the timer c) After checking the origin sensor, it does an emergency stop. Please then go home return operation

Error to return at the origin return operation (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Socket side collet down error	Rotary head can not move because the pocket side collet is lowering.			After an emergency stop, please Returns to origin
Pocket motor operationr timeout error	Motor operation of the pocket side collet has timed out			a) Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
Collet A up timeout error	Upward movement of the collet A has timed out			a)Check the obstacles that are preventing the upward movement b) Checking the motor parameters c) Checking the timer d) After checking the origin sensor, it does an emergency stop. Please then go home return operation
Collet A down timeout error	Downward movement of the collet A has timed out			a) Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
Collet B up timeout error	Upward movement of the collet B has timed out			a)Check the obstacles that are preventing the upward movement b) Checking the motor parameters c) Checking the timer d) After checking the origin sensor, it does an emergency stop. Please then go home return operation
Collet B down timeout error	Downward movement of the collet A has timed out			a) Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
Collet A XYT motor timeout error.	It was operation time out of the XY theta motor of the collet A.			a)Check of the collet XY theta offset b)Checking the motor parameters c) Checking the timer d) After an emergency stop, please Returns to origin

Error to return at the origin return operation (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Collet B XYT motor timeout error.	It was operation time out of the XY theta motor of the collet B.			a)Check of the collet XY theta offset b)Checking the motor parameters c) Checking the timer d) After an emergency stop, please Returns to origin
Collet A XY motor timeout error.	operation time-out error of the XY motor of the collet A has occurred.			a)Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
Collet B XY motor timeout error.	operation time-out error of the XY motor of the collet B has occurred.			a)Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
Collet A Theta motor timeout error.	operation time-out error of theta motor of the collet A has occurred.			a)Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
Collet B Theta motor timeout error.	operation time-out error of theta motor of the collet B has occurred.			a)Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
Rotary head motor operation timeout error	operation time-out error of the rotary head has occurred.			a)Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
Pocket motor operationr timeout error	operation time-out error of rotary pocket has occurred.			a)Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
Robot hand down error	Rotary pocket can not move because the hand is lowering			After an emergency stop, please Returns to origin
Pocket side collet motor operation timeout error	Motor operation of the pocket side collet has timed out. (Manual operation)			a)Check the obstacles that are preventing the operation. b) After an emergency stop, please Returns to origin

Error to return at the origin return operation (Continuation)

Error name	Cause	Input IO	Normal status	Release method
Socket side collet down error	Rotary head can not move because the socket side collet is lowering.			After an emergency stop, please Returns to origin
Socket side collet motor operation timeout error	operation time-out error of the socket side collet has occurred.			a)Move the unit that is preventing the operation b)Checking the motor parameters c) Checking the timer d) After an emergency stop, please Returns to origin
LD unit origin adjustment timeout error.	Origin return time-out error of the LD unit has occurred.			a)Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
LD unit adjustment timeout error.	operation time-out error of the LD unit has occurred.			a)Checking the motor parameters b) Checking the timer c) After an emergency stop, please Returns to origin
LD unit position error.	Since the position of the LD unit is inappropriate, it is not possible to move the rotary head.			a)Check the position data b) After an emergency stop, please Returns to origin
Socket clamp adjustment timeout error.	Socket positioning operation has timed out.			a) Checking the timer b) After an emergency stop, please Returns to origin
Tray3 timeout error	The operation of the tray 3 has timed out.			a) Checking the timer b) After an emergency stop, please Returns to origin
Tray3 Forward timeout error	Forward movement of the tray 3 has timed out.	Tray3 front side limit Tray3 back side limit	OFF ON	a)Sensor status check b) Checking the timer c) After an emergency stop, please Returns to origin
Tray3 Backward timeout error	Backward movement of the tray 3 has timed out.	Tray3 front side limit Tray3 back side limit	ON OFF	a)Sensor status check b) Checking the timer c) After an emergency stop, please Returns to origin

4.2 Operator call

4.2.1 Processing at the time of operator call occurs

For some reason, it will be displayed when the operator occurs operator call stop screen. If the operator call screen is displayed, after correctly processing the cause, and then close the operator call screen by pressing the reset button.. It lists the operator call to the next section.

Fig 4.2A Operator call screen

4.2.2 Operator call

Error name	Cause	Input IO	Normal status	Release method
Empty tray buffer full error	Empty tray buffer is full.	Tray transfer landing check	ON	Remove the one-sheet tray from the empty tray buffer
Receiving tray absence error	There is no tray to empty tray buffer. Or it failed to adsorption of the empty tray.			Add the tray to empty tray buffer.
Pretest tray full error	It has become the supply tray full of Pretest tray	Pretest tray elevator level	OFF	Remove the Pretest tray
Pretest tray absence error	Pretest tray does not have.			a)Add the tray to Pretest tray b)if you do not want to continue the construction, press the lot end button
Please change the pretest tray.	Pretest tray of the product is now empty. Please Remove the empty tray.			Remove the empty tray of Pretest tray
Tray 1 full error	Storage tray of the tray 1 is full	Tray1 elevator level	OFF	Remove the tray of the tray 1
TRAY 1 FULL	Store the number of tray 1 becomes full, it was stopped.			After checking the storage state of the tray 1, and re-start.
Tray 1 full stop	Storage tray number of the tray 1 warned stop reached the designated number			Please re-start to remove the storage tray of the tray 1.
Tray 2 full error	Storage tray of the tray 2 is full	Tray2 elevator level	OFF	Remove the tray of the tray 2
TRAY 2 FULL	Store the number of tray 2 becomes full, it was stopped.			After checking the storage state of the tray 2, and re-start.
Please add a empty tray at the tray 2.	Store the number of tray 2 becomes full, it was stopped.			After checking the storage state of the tray 2, please re-start by adding an empty tray.
Please add a empty tray at the tray 2.	When the tray 2 is specified in the empty tray buffer, there is no empty tray.			Add the empty tray to tray 2.
Tray 3 absence error	Tray is not in the tray 3.			Add the empty tray to tray 3.
Tray 3 direction error	Tray direction of the tray 3 is reversed.			The direction of the tray after checking, set the empty tray in the correct direction

Operator call(Continuation)

Error name	Cause	Input IO	Normal status	Release method
Replace tray 3.	Store the number of the tray 3 becomes full, it was stopped.			Add the empty tray remove the tray 3.
Please change the tray 4	Store the number of the tray 4 becomes full, it was stopped.			Add the empty tray remove the tray 4.
Please change the tray 5	Store the number of the tray 5 becomes full, it was stopped.			Add the empty tray remove the tray 5.
Please change the tray 6	Store the number of the tray 6 becomes full, it was stopped.			Add the empty tray remove the tray 6.
Please change the tray 7	Store the number of the tray 7 becomes full, it was stopped.			Add the empty tray remove the tray 7.
Please change the tray 8	Store the number of the tray 8 becomes full, it was stopped.			Add the empty tray remove the tray 7.
Cycle stop	At the specified location, it was cycle stop.			After checking the stopped state, and re-start.
Please clean the socket.	Contact number of the socket has reached the specified number of times.			After you clean the socket, and re-start